	FIR VEHICLE TESTING	19/07/2023
FIR No :FIR/Testing-Outdoor/224000/2023/6316		
Part No/ Group :22404168R		
Subject :Performance evaluation of Signa 4221.T Cx RDE in comparison with Ashok Leyland 4220.T BSVI and Signa 4221.T BSVI base vehicle.		
Test Vehicle	Project Name :Signa 4221.T Cx RDE Model Name : Signa 4221.T Cx RDE Chasis Number :PC44709022RJ00810 Engine Number :HXX104198	Enclosures : Annexure 1 - 6
FIR Status :	ESO FIR	

Role	Name	Signature
Prepared By	Palash Das	19-Jul-2023
Checked By	Swapnil Jadhav	19-Jul-2023
Approved By	Ajaykumar Jadhav	19-Jul-2023

Cc: Head CVBU Engg, Chief Engineer, Head COC, Project Team [Design]; Project Team [Testing]

PA 20 - F 222 / May19

Documents : Please click here

FIR_Testing_Outdoor_22400..

Subject: -	Performance evaluation of Signa 4221.T Cx RDE in comparison with Ashok Leyland 4220 BSVI and Signa 4221.T BSVI base vehicle.		

Conclusion:

➢ **Signa 4221.T Cx RDE in comparison with Ashok Leyland 4220 BSVI**

- Overall performance of Signa 4221.T Cx BSVI RDE is better as compared to Ashok Leyland 4220 BSVI HM PALS position “C” achieved against “C” — Action by: -- — Failure class: --

➢ **Signa 4221.T Cx RDE in comparison with Signa 4221.T BSVI**

- Overall performance of Signa 4221.T Cx BSVI RDE is better as compared to Signa 4221.T BSVI base vehicle. — Action by: -- — Failure class: --

1.1 Acceleration:

➢ **Signa 4221.T Cx RDE in comparison with Ashok Leyland 4220 BSVI**

- Acceleration (0-60 kmph) of Signa 4221.T Cx BSVI RDE was found to be **72.4** Sec. Signa 4221.T BSVI RDE is **8.79%** better as compared to Ashok Leyland 4220 BSVI vehicle and meets PALS target. — Action by: -- — Failure class: --
- Acceleration (0-1000m) of Signa 4221.T Cx BSVI RDE was found to be **84** Sec. Signa 4221.T Cx BSVI RDE is **4.86%** better as compared to Ashok Leyland 4220 BSVI vehicle and meets PALS target. — Action by: -- — Failure class: --

➢ **Signa 4221.T Cx RDE in comparison with Signa 4221.T BSVI**

- Acceleration (0-60 kmph) of Signa 4221.T Cx BSVI RDE is **Superior by 8.93%** as compared to Signa 4221.T BSVI vehicle. — Action by: -- — Failure class: --
- Acceleration (0-1000m) of Signa 4221.T Cx BSVI RDE is **Superior by 3.93%** as compared to Signa 4221.T BSVI vehicle. — Action by: -- — Failure class: --

1.2 Wide open throttle drivability:

➢ **Signa 4221.T Cx RDE in comparison with Ashok Leyland 4220 BSVI**

- In gear drivability of Signa 4221.T Cx BSVI RDE vehicle was found to be **1.82** kmph/s, **2.01** kmph/s, **2.19** kmph/s, **1.86** kmph/s, **1.54** kmph/s, **0.91** kmph/s, **0.80** kmph/s & **0.50** kmph/s in 1st ,2nd,3rd ,4th ,5th, 6th ,7th & 8th gear respectively. — Action by: -- — Failure class: --

	Action by	Failure class
• Rate of acceleration of Signa 4221.T Cx BSVI RDE vehicle is inferior to AL 4220 BSVI in 1st, 2nd, 3rd Gears.	VI/ENG	C
1.3 Bathtub drivability:		
➢ **Signa 4221.T RDE in comparison with Ashok Leyland 4220 BSVI**		
• Bathtub drivability of Signa 4221.T Cx BSVI RDE vehicle was found **better** as compared to Ashok Leyland 4220 BSVI vehicle in all gears except 5th gear (10 - 15 Kmph) & 7th gear (20 - 25 Kmph). Vehicle meeting PALS target in all gears except 5th & 7th Gear.	VI/ENG	C
➢ **Signa 4221.T BSVI RDE in comparison with Signa 4221.T BSVI**		
• Bathtub drivability of Signa 4221.T Cx BSVI RDE vehicle was found **superior** to Signa 4221.T BSVI vehicle in all gears.	--	--
1.4 Sustain and Restart Gradability:		
➢ **Signa 4221.T Cx BSVI RDE in comparison with Ashok Leyland 4220 BSVI**		
• Sustain gradeability of Signa 4221.T Cx BSVI RDE Veh. Was found to be **29.82%.** Signa 4221.T Cx BSVI RDE Veh is **7.34%** superior as compared to Ashok Leyland 4220.T BSVI vehicle.	--	--
• Restart gradeability of Signa 4221.T Cx BSVI RDE was found to be **25.72%.** Signa 4221.T Cx BSVI RDE is **6.02%** superior as compared to Ashok Leyland 4220.T BSVI vehicle.	--	--
➢ **Signa 4221.T Cx BSVI RDE in comparison with Signa 4221.T BSVI**		
➢ Sustain gradeability of Signa 4221.T Cx BSVI RDE Veh. is **At par** as compared to Signa 4221.T BSVI base vehicle.	--	--
➢ Restart gradeability of Signa 4221.T Cx BSVI RDE Veh. is **At par** as compared to Signa 4221.T BSVI base vehicle.	--	--
2.0 Result: Wide observation.		
FIR Status	**Page 2 of 5**	

FAILURES CLASS:

A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement.

B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention.

C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule.

D: Customer perception issues and minor irritants.

3.0 Test Vehicle:

1. Signa 4221.T Cx BSVI RDE (ERC – 1253)
   Chassis No. – PC44709022RJ00810
   Engine No. – HXX104198

Input Doc Ref. : Testing-Outdoor/15688

DID No : NA

DDP No : NA

Enclosures:

Annexure 1: Panel chart

Annexure 2: PAT Sheet

Annexure 3: PAT Summary

Annexure 4: Test request

Annexure 5: Test plan

Work Order No. :- P.21.EA.458 | **Make :-** Tata Motors Ltd | **T I No. – 42546**

Earlier Ref: -

- FIR/Testing-Outdoor/224000/2021/3650 Report on performance evaluation of Signa 4221.T 5L BSVI vehicle in Comparison with Ashok Leyland 4220 BSVI vehicle Prima 4225.T Atulya Vehicle.

4.0 Project Status: DR3

5.0 Introduction:

- As a part of RDE program, Signa 4221.T Cx BSVI RDE vehicle was offered for performance evaluation, Ashok Leyland 4220 BSVI was taken as core benchmark and Signa 4221.T BSVI vehicle was taken as Base for performance comparison. This report will reveal detail of this trial with the status.

FIR Status

6.0 Testing

➢ Following test was conducted during trials.

Sr. No.	Test Description	Standard Referred	Instrument used
1	Acceleration	TST_VT_WG_0002	VBOX SX with GPS, Weighbridge (CRS Gate)
2	WOT Drivability	TST_VT_WG_0002	
3	Bath -Tub Drivability	TST_VT_WG_0261	
4	Gradability	TST_VT_WG_11	
5	Restart Gradability	TST_VT_WG_127	
6	Coastdown	TST_VT_WG_36	

7.0 Vehicle details:

Sr. No.	Parameters			Base Vehicle : OBD-1	Core Benchmark	Test Vehicle : OBD-2
				SIGNA 4221.T BSVI	AL 4220.T BSVI	SIGNA 4221.T RDE BSVI
1	Vehicle Identification No. (ERC No. /EJ No./Reg. No.)			ERC-1153	WB 83 4133	ERC-1253
2	Vehicle GVW/GCW		kg	42000	42000	42000
3	Engine Details (Make/cc/Norms/no. of Cyl.)			TML 5LNGE	H-Series 5.7L CRS BSVI	TML 5LNGE
	Engine Max Power			200 HP @ 2200rpm	200 hp @ 2400 rpm	204 HP @ 2200rpm
	Engine Max Torque			850Nm @ 1000-1600rpm	700 Nm @ 1200-2000 rpm	850Nm @ 1000-1600rpm
	Engine Calibration Maturity Level			As per production released	As per production released	95% Cal Maturity
4	Regeneration Strategy (Time ase/Soot base)			Time Based (110 Hrs)	--	Time Based (110 Hrs)
5	Gear Box (make/drive/FGR)			G1150-FGR - 9.13,DD	ZF 8S-1110 Gearbox	G1150-FGR - 9.13,DD
	Gear Ratios			1-9.13, 2- 6.72, 3- 4.90, 4- 3.57, 5- 2.55, 6- 1.88, 7- 1.37, 8- 1.00, Rev: 13.40, Cr:12.87	1st: 8.83, 2nd: 6.28, 3rd: 4.64, 4th: 3.48, 5th: 2.54, 6th: 1.81, 7th: 1.34, 8th: 1, Rev:12.04, Cr:12.73	1-9.13, 2- 6.72, 3- 4.90, 4- 3.57, 5- 2.55, 6- 1.88, 7- 1.37, 8- 1.00, Rev: 13.40, Cr:12.87
6	RAR			5.85	6.17	5.85
7	Tyre Details			All 295/90R 20 (JK Make)	Apollo 295/90R 20	All 295/90R 20 (JK Make)
8	LBSC	High & Low Engine Speed Breakpoint	RPM	--	--	--
		Vehicle Mass Threshold	kg	--	--	--
	VAM	Variable Acceleration 1, 2	kmph/sec	--	--	--
		Variable Acceleration Speed 1, 2	kmph	--	--	--
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)	kmph	--	--	--
	RSG	Maximum Accelerator Vehicle Speed	kmph	80	80	80
	Torque Factor	Torque Limit Medium Adjustment Factor	Balance	--	--	--
		Torque Limit Maximum Adjustment Factor	Economy	--	--	--
	PTP	Powertrain Protection (Enable / Disable)		Disable	Disable	Disable
		Torque Limit	Nm	NA	NA	NA
	USFE / Torque Table Revision (1/2/3/n)			--	--	--
9	Fan Details (type/make/blades)			24" E- Viscous fan	--	24" E- Viscous fan
10	Base Cal Improvement (Cal Maturity %)			As per production released	As per production released	95% Cal Maturity

Vehicle Photographs

8.0 Torque table:-

8.3.1 : Base vehicle (Signa 4221.T BS-VI OBD-I (200 HP))

FE Switch Torque Table (Torque Limit)										
Mode	Load Gear	1st Gear	2nd Gear	3rd Gear	4th Gear	5th Gear	6th Gear	7th Gear	8th Gear	Reverse
Economy Mode	850	850	850	700	590	590	590	590	590	850
Balance Mode	850	850	850	700	700	700	700	700	700	850
Power Mode	850	850	850	850	850	850	850	850	850	850

8.3.2: Test vehicle (Signa 4221.T BS-VI OBD-II (204 HP))

FE Switch Torque Table (Torque Limit)										
Mode	Load Gear	1st Gear	2nd Gear	3rd Gear	4th Gear	5th Gear	6th Gear	7th Gear	8th Gear	Reverse
Economy Mode	850	850	850	700	590	590	590	590	590	850
Balance Mode	850	850	850	700	700	700	700	700	700	850
Power Mode	850	850	850	850	850	850	850	850	850	850

9.0 Observation:-

Overall performance Signa 4221.T Cx BSVI RDE is better as compared to Ashok Leyland 4220 BSVI and PALS position achieved “C” against “C”.

10. Future testing: - As per further development.

Format No. PA20 F222 (R4) March, 2021

Level 1 Attribute:

PAT- Performance and Drivability

PANEL chart

VE CVBU/ PD/001

Specification:
Max Power-204 hp @ 2200 rpm
Max Torque - 850 Nm 1000-1600 rpm
Driveline : G-950 DD | 5.85 RAR | 295/90R 20 Tyres | 3 Mode FE Switch | Torque Factor: NA
EOL Parameters : RSG: 80 kmph

Panel Chart No.: VE CVBU/ HCV/ Signa 4221.T Cx BSVI RDE 5L/ PD/001

Programme / MY : SIGNA 4221.T Cx 5L BSVI RDE	**Gateway :** DR3	**Owner: Ajaykumar Jadhav**	**Project Name: SIGNA 4221.T 5L BSVI RDE(Cx Version)**	**Date:** 17/07/2023

Level 1 Summary:

		PALS	Status: Current/ Gateway	Status: Job#1 Predicted
PALS	Level 1:	C	C	

Statement Of Intent (SOI) or DNA Description:

Performance: Haulage application

Attribute Target Summary:

Level 2 / Level 3 / Level 4 / Level 5 / Level 6	PALS	Metric	Current Target w.r.t Ashok Leyland 4220.T BSVI	Status: Current/Gateway SIGNA 4221.T Cx BSVI RDE	Base Vehicle: Signa 4221.T BSVI	Competitor Data: Core Benchmark Ashok Leyland 4220	Remarks
				Heavy mode	**Heavy mode**	--	
Specification				**LBSC - NA VAM- NA GDP- NA RSG - 80 kmph**	**LBSC - NA VAM- NA GDP- NA RSG - 80 kmph**	**--**	--
Vehicle GVW (Payload)		kg		42000	42000	42000	
Power (Max) @ rpm		HP		204 hp @ 2200 rpm	200 hp @ 2200 rpm	200 hp @ 2400 RPM	
Power to Weight ratio		HP / Ton		4.85	4.76	4.76	
Torque (Max) @ rpm		Nm		850 Nm @ 1000-1600 rpm	850 Nm @ 1000-1600 rpm	700 Nm at 1200-2000 rpm	
Torque to Weight ratio		Nm / Ton		20.24	20.24	16.67	
Tyres size		~		JK 295/90R20	JK 295/90 R20	Apollo Tyres 295/90 R20	
Gear Box Ratios		~		G1150 DD, FGR -9.13	G1150 DD, FGR -9.13	Eaton DD 9S Gearbox	
Final Drive Ratio		~		**5.85**	**5.85**	**6.17**	
Acceleration at rated load	C			C			
0-60 kmph		Sec	**≤ 79.38**	**72.40**	79.5	79.38	
0-1000 m		Sec	**≤ 88.29**	**84.00**	87.44	88.29	
Bathtub Drivability (Time) at Rated load	C			C			
3rd Gear							
5 - 10 Kmph		sec	**≤ 3.34**	**3.34**	--	3.34	
10 - 15 Kmph		sec	**≤ 2.75**	**1.98**	2.59	2.75	
4th Gear							
10 - 15 Kmph		sec	**≤ 3.68**	**3.13**	3.27	3.68	
15 - 20 Kmph		sec	**≤ 2.90**	**2.51**	2.68	2.90	
5th Gear							
10 - 15 Kmph		sec	**≤ 4.79**	**5.75**	5.54	4.79	
15 - 20 Kmph		sec	**≤ 4.01**	**3.36**	3.80	4.01	
20-25 kmph		sec	**≤ 3.88**	**3.22**	4.18	3.88	
25-30 kmph		sec	**≤ 4.41**	**4.06**	5.03	4.41	
6th Gear							
20-30 kmph		sec	**≤ 11.71**	**8.75**	10.17	11.71	
30-40 kmph		sec	**≤ 11.96**	**10.29**	12.02	11.96	
7th Gear							
20-25 kmph		sec	**≤ 9.33**	**10.36**	10.61	9.33	
25-30 kmph		sec	**≤ 8.34**	**6.59**	6.50	8.34	
30-35 kmph		sec	**≤ 7.97**	**5.53**	6.78	7.97	
35-40 kmph		sec	**≤ 7.74**	**6.24**	6.74	7.74	
40-45 kmph		sec	**≤ 7.86**	**6.27**	7.87	7.86	
45-50 kmph		sec	**≤ 8.89**	**7.46**	9.03	8.89	
8th Gear							
30-40 kmph		sec	**≤ 24.46**	**22.64**	25.38	24.46	
40-50 kmph		sec	**≤ 25.44**	**17.12**	21.09	25.44	
50-60 kmph		sec	**≤ 26.08**	**20.21**	24.13	26.08	
60-70 kmph		sec	**≤ 32.36**	**25.74**	34.12	32.36	
Gradeability at rated load	C			C			
Sustain Gradeability		%	**≥ 27.78**	**29.82**	29.82	27.78	--
Restart Gradeability		%	**≥ 24.26**	**25.72**	25.72	24.26	

Remarks for the Bathtub Drivability rows: --

Executive Summary:
1. Ashok Leyland 4220.T BSVI have been taken for target setting.
2. Signa 4221.T Cx BSVI RDE was having 95% engine calibration maturation.
3.Target has been taken at-par as per Cx variant approach.
4. Sustain & Restart Gradeability was extraploted from base vehicle with same configuration.

Tested by:	**Verified and Approved by:**
Palash Das / Swapnil Jadhav PAT Engineer	Ajaykumar Jadhav / Sanjeev Kumar PAT Lead / Project Lead

TATA

TATA MOTORS LTD.
ERC,PUNE

PAT ATTRIBUTES SHEET
Performance and Drivability | Vehicle-level Engine Calibration
Features / End of Line | OBDII and Diagnostics

Document Version: VE CVBU/PD/001

Document Revision No.: VE CVBU/HCV/ Signa 4221.T Cx RDE/ 001

Department Name: Vehicle Engineering

Project Name: Signa 4221.T 5L BSVI_Cx RDE
Base Vehicle- SIGNA 4221.T 5L BSVI_G1150 9S DD (FGR- 9.13)_RAR 5.85_295/90R20 tires_42 Ton GVW_3 Mode FE Switches
Competitor Vehicle- Ashok Leyland 4220 BSVI_ZF 9S-1110 DD (FGR- 8.83)_RAR 6.17_295/90R20 tires_42 Ton GVW
Application- Haulage

Date: 17/07/2023

Sr.No.	Aggregate and test parameter: Level 2	Level 3	Level 4	Level 5	Level 6	Level 7	Unit	Test Type	Classification @DR3	Ref.Standard / Work Guideline	Simulation	Base Vehicle: Signa 4221.T 5L BSVI G1150 DD 9S RAR 5.85	Core Benchmark: Ashok Leyland 4220 9 Speed DD RAR 6.17	Target	Test Vehicle: Signa 4221.T Cx RDE BSVI G1150 DD 9S RAR 5.85	Status (RYG)	Remarks	CoC Remark
A	Performance Evaluation								Heading			--	--	--	--	--	--	
1.2.9			Vehicle maximum speed				Kmph	Standard	Specification	TST/VT/WG/02		80.0	80.0	80	80	G	--	
1.4		Acceleration							Heading							--		
1.4.2			AC Off						Heading							--		
1.4.2.1			0-60 kmph (0-37 mph)				Sec	Standard	Target	TST/VT/WG/02		79.50	79.38	≤ 79.38	72.40	G		
1.4.2.4			0-1000 m				Sec		Target			87.44	88.29	≤ 88.29	84.00	G		
1.5		Grad ability							Heading							--		
1.5.2			AC Off						Heading							--		
1.5.2.1			Maximum (sustained) Gradeability (forward)				%	Standard	Target	TST/VT/WG/011		29.82	27.78	≥ 27.78	29.82	G	Sustain & Restart Gradeability was extrapolted from base vehicle with same configuration.	
1.5.2.2			Max Restart Gradeability (forward)				%	Standard	Target	TST/VT/WG/127		25.72	24.26	≥ 24.26	25.72	G		
B	Drivability Evaluation								Heading							--		
1.3		WOT Drivability (AC Off)							Heading							--		
1.3.1			1st gear 4-6 Kmph				Sec		Data Generation			1.10	0.99	--	1.10	--		
1.3.2			2nd gear 6-9 Kmph				Sec		Data Generation			1.38	1.39	--	1.49	--		
1.3.3			3rd gear 8- 14 Kmph				Sec		Data Generation			2.67	2.72	--	2.74	--		
1.3.4			4th gear 10- 18 Kmph				Sec		Data Generation			4.40	4.43	--	4.31	--		
1.3.5			5th gear 15 - 25 Kmph				Sec		Data Generation			6.73	7.39	--	6.48	--	--	
1.3.6			6th gear 16 - 24 kmph				Sec		Data Generation			8.31	8.94	--	8.81	--		
1.3.7			7th gear 25 - 35 kmph				Sec		Data Generation			12.69	16.08	--	12.46	--		
1.3.8			8th gear 30 - 45 kmph				Sec		Data Generation			31.57	37.05	--	30.05	--		
1.3.10		AC Off (Rate of acceleration)						Standard	Heading	TST/VT/WG/02						--		
1.3.11			1st gear 4-6 Kmph				Kmph/sec		Data Generation			1.82	2.03	--	1.82	--		
1.3.12			2nd gear 6-9 Kmph				Kmph/sec		Data Generation			2.17	2.16	--	2.01	--		
1.3.13			3rd gear 8- 14 Kmph				Kmph/sec		Data Generation			2.25	2.21	--	2.19	--		
1.3.14			4th gear 10- 18 Kmph				Kmph/sec		Data Generation			1.82	1.81	--	1.86	--		
1.3.15			5th gear 15 - 25 Kmph				Kmph/sec		Data Generation			1.49	1.35	--	1.54	--	--	
1.3.16			6th gear 16 - 24 kmph				Kmph/sec		Data Generation			0.96	0.90	--	0.91	--		
1.3.17			7th gear 25 - 35 kmph				Kmph/sec		Data Generation			0.79	0.62	--	0.80	--		
1.3.18			8th gear 30 - 45 kmph				Kmph/sec		Data Generation			0.48	0.40	--	0.50	--		
1.4		Bathtub Curve - Part Throttle Drivability (AC Off)							Heading							--		
1.4.13			3rd gear						Heading									
1.4.14			5 - 10 Kmph				sec		Target			--	3.34	≤ 3.34	3.34	G		
1.4.15			10 - 15 Kmph				sec		Target			2.59	2.75	≤ 2.75	1.98	G		
1.4.16			4th gear				sec		Heading							--		
1.4.16			10 - 15 Kmph				sec		Target			3.27	3.68	≤ 3.68	3.13	G		
1.4.17			15-20 kmph				sec		Target			2.68	2.90	≤ 2.9	2.51	G		
1.4.1			5th gear						Heading									
1.4.4			10 - 15 Kmph				sec		Target			5.54	4.79	≤ 4.79	5.75	R		
1.4.5			15 - 20 Kmph				sec		Target			3.80	4.01	≤ 4.01	3.36	G		
1.4.6			20-25 kmph				sec		Target			4.18	3.88	≤ 3.88	3.22	G		
1.4.8			25-30 kmph						Target			5.03	4.41	≤ 4.41	4.06	G	1. Ashok Leyland 4220.T BSVI have been taken for target setting. 2. Signa 4221.T Cx BSVI RDE was having 95% engine calibration maturation. 3.Target has been taken at-par as per Cx variant approach	
1.4.9			6th gear				sec		Heading							--		
1.4.10			20-30 kmph				sec	Standard	Target	TST/VT/WG/261		10.17	11.71	≤ 11.71	8.75	G		
1.4.11			30-40 kmph				sec		Target			12.02	11.96	≤ 11.96	10.29	G		
1.4.12			7th gear				sec		Heading							--		
1.4.12			20-25 kmph				sec		Target			10.61	9.33	≤ 9.33	10.36	R		
1.4.12			25-30 kmph				sec		Target			6.50	8.34	≤ 8.34	6.59	G		
1.4.13			30-35 kmph						Target			6.78	7.97	≤ 7.97	5.53	G		
1.4.14			35-40 kmph				sec		Target			6.74	7.74	≤ 7.74	6.24	G		
1.4.9			40-45 kmph				sec		Target			7.87	7.86	≤ 7.86	6.27	G		
1.4.15			45-50 kmph				sec		Target			9.03	8.89	≤ 8.89	7.46	G		
1.4.26			8th gear						Heading									
1.4.27			30-40 Kmph				sec		Target			25.38	24.46	≤ 24.46	22.64	G		
1.4.28			40-50 Kmph				sec		Target			21.09	25.44	≤ 25.44	17.12	G		
1.4.46			50-60 Kmph				sec		Target			24.13	26.08	≤ 26.08	20.21	G		
1.4.47			60-70 Kmph				sec		Target			34.12	32.36	≤ 32.36	25.74	G		
C	Engine Calibration Activities								Heading							--		
1.0		System Verification Trials							Heading							--		
1.1			Powertrain Characteristics (AC ON)						Target				--	--	--	NA	--	
1.2.1			Powertrain Characteristics (AC OFF)						Target				--	Pass	Pass	G	--	
2.0		Vehicle Perfomance Trials							Heading							--		
2.1			Vehicle Speed detection						Target				--	Pass	Pass	G	--	
2.2			Engine Speed detection						Target				--	Pass	Pass	G	--	
2.3			Gear Detection						Target				--	Pass	Pass	G	--	
2.4			Driven Idle						Target				--	Pass	Pass	G	--	
3.0		Engine derate verification							Heading							--		
3.2			Limp home validation						Heading				--	Pass	Pass	G	--	
3.3			Reduced torque fault validation						Heading				--	Pass	Pass	G	--	
4.0		Evaluation of Take-off ability, Acceleration and Deceleration							Target					Pass		G		
D	Vehicle feature validation and Fuel Economy features / End of Line (EOL) parameters segreggation and verification								Heading							--		
1.0		Vehicle Mass Estimation							Target				--	Pass	Pass	NA	--	
2.0		Road Slope Estimation							Target				--	Pass	Pass	NA	--	
3.0		Vehicle acceleration management (VAM tuning)							Target				--	Pass	Pass	G	--	
4.0		Electronic vehicle speed Limiter (EVSL) / Load based speed control (LBSC)							Target				--	Pass	Pass	NA	--	
5.0		Vehicle speed limiter (Top and one-gear down)							Target				--	Pass	Pass	NA	--	
6.0		EVSL+ VAM / LBSC+VAM							Target				--	Pass	Pass	NA	--	
7.0		Gear Down Protection (GDP)							Target				--	Pass	Pass	NA	--	
8.0		Idle shutdown							Target				--	Pass	Pass	G	--	
9.0		Average fuel economy (AFE)							Target				--	Pass	Pass	G	--	
10.0		IFE							Target				--	Pass	Pass	NA	--	
11.0		Engine Brake control							Target				--	Pass	Pass	G	--	
12.0		Cruise control (CC)							Target				--	Pass	Pass	G	--	
13.0		Power Take off (PTO)							Target				--	Pass	Pass	NA	--	
14.0		Remote PTO							Target				--	Pass	Pass	NA	--	
15.0		Road Speed governor							Target				--	Pass	Pass	G	--	
16.0		Engine Protection shutdown							Target				--	Pass	Pass	G	--	
17.0		Powertrain protection							Target				--	Pass	Pass	NA	--	
18.0		Trip-Info							Target				--	Pass	Pass	G	--	
E	Environmental Trials Verification								Heading							--		
1.0		Summer trials							Heading							--		
1.4			Hot start						Target				--	Pass	Pass	G		
1.5			Low idle governor calibration						Target				--	Pass	Pass	G	Carry over from OBD I validation	
1.6			Driven Idle						Target				--	Pass	Pass	G		
1.7			Subjective Driveability Assessment						Target				--	Pass	Pass	G		
2.0		Winter trials							Heading							--		
2.2			Cold Start (at Ambient as per Application requirement)						Target				--	Pass	Pass	G		
2.3			Smoke level (visually) during free acceleration after start						Target				--	Pass	Pass	G	Carry over from OBD I validation	
2.4			Subjective Driveability Assessment						Target				--	Pass	Pass	G		
6.0		High altitude trials/ calibration							Heading							--		
6.2			Low idle governor calibration				rpm	New Devlopment	Target				--	Pass	Pass	G		
6.9.1			%Deterioration in Drivability with base calibration				%		Target				--	Pass	Pass	G		
6.10.1			Turbocharger duty cycle mapping						Target				--	Pass	Pass	G	Carry over from OBD I validation	
6.10.1.1			Smoke during driving present or not						Target				--	Pass	Pass	G		
6.10.1.2			Assessment of turbocharger Whistling noise					New Devlopment	Target				--	Pass	Pass	G		
6.11.1			Power calculation				%		Target				--	Pass	Pass	G		
F	Onboard Diagnostics								Heading							--		
5.0		OBD Diagnostic - TML tool verification							Heading				--	Pass	Pass	G		
7.0		Derate Strategies for Critical Faults							Heading				--	Pass	Pass	G	Carry over from OBD I validation	
8.0		Verification of OBD interface for Comprehensive Component Monitors							Heading				--	Pass	Pass	G		

Sign Off			Index for Status
Responsibility	Name	Signature	Off target, no resolution
Prepared by: PAT Engineer	Palash Das/Swapnil		Marginally off target, resolution planned/low risk
Checked & Approved by: PAT Lead / Project Lead	Ajaykumar Jadhav / Sanjeev kumar		Met target, no concern

PED evaluation of Signa 4221.T RDE BSVI in comparison with Ashok Leyland 4220 BSVI and Signa 4221.T 5L BSVI Vehicle

PED| 17.07.23

OUR CULTURE CREDO

AT TATA MOTORS
We are connecting aspirations by being bold in thought and action, owning every opportunity and challenge, Solving together as one team and engaging all our stakeholders with empathy.
We are **MORE WHEN ONE!**

Content

TATA MOTORS
Connecting Aspirations

- ✓ Vehicle specification comparison
- ✓ Performance Data Summary
- ✓ Minimum and Maximum Gear Speed
- ✓ Wide open throttle performance
- ✓ Tractive effort comparison
- ✓ Acceleration
- ✓ WOT drivability
- ✓ Bathtub drivability
- ✓ Coast down comparison
- ✓ Sustain and restart Gradeability

Vehicle Specification Comparison

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Sr. No.	Parameters		Unit	Base Vehicle : OBD-1	Core Benchmark	Test Vehicle : OBD-2
				SIGNA 4221.T BSVI	AL 4220.T BSVI	SIGNA 4221.T RDE BSVI
1	Vehicle Identification No. (ERC No. /EJ No./Reg. No.)			ERC-1153	WB 83 4133	ERC-1253
2	Vehicle GVW/GCW		kg	42000	42000	42000
3	Engine Details (Make/cc/Norms/no. of Cyl.)			TML 5LNGE	H-Series 5.7L CRS BSVI	TML 5LNGE
	Engine Max Power			200 HP @ 2200 rpm	200 hp @ 2400 rpm	204 HP @ 2200 rpm
	Engine Max Torque			850Nm @ 1000-1600 rpm	700 Nm @ 1200-2000 rpm	850Nm @ 1000-1600rpm
	Engine Calibration Maturity Level			As per production released	As per production released	95% Cal Maturity
4	Regeneration Strategy (Time ase/Soot base)			Time Based (110 Hrs)	--	Time Based (110 Hrs)
5	Gear Box (make/drive/FGR)			G1150-FGR - 9.13,DD	ZF 8S-1110 Gearbox	G1150-FGR - 9.13,DD
	Gear Ratios			1-9.13, 2- 6.72, 3- 4.90, 4- 3.57, 5- 2.55, 6- 1.88, 7- 1.37, 8- 1.00, Rev: 13.40, Cr:12.87	1st: 8.83, 2nd: 6.28, 3rd: 4.64, 4th: 3.48, 5th: 2.54, 6th: 1.81, 7th : 1.34, 8th: 1, Rev:12.04, Cr:12.73	1-9.13, 2- 6.72, 3- 4.90, 4- 3.57, 5- 2.55, 6- 1.88, 7- 1.37, 8- 1.00, Rev: 13.40, Cr:12.87
6	RAR			5.85	6.17	5.85
7	Tyre Details			All 295/90R 20 (JK Make)	Apollo 295/90R 20	All 295/90R 20 (JK Make)
8	LBSC	High & Low Engine Speed Breakpoint	RPM	--	--	--
		Vehicle Mass Threshold	kg	--	--	--
	VAM	Variable Acceleration 1, 2	kmph/sec	--	--	--
		Variable Acceleration Speed 1, 2	kmph	--	--	--
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)	kmph	--	--	--
	RSG	Maximum Accelerator Vehicle Speed	kmph	80	80	80
	Torque Factor	Torque Limit Medium Adjustment Factor	Balance	--	--	--
		Torque Limit Maximum Adjustment Factor	Economy	--	--	--
	PTP	Powertrain Protection (Enable / Disable)		Disable	Disable	Disable
		Torque Limit	Nm	NA	NA	NA
	USFE / Torque Table Revision (1/2/3/n)			--	--	--
9	Fan Details (type/make/blades)			24'' E- Viscous fan	--	24'' E- Viscous fan
10	Base Cal Improvement (Cal Maturity %)			As per production released	As per production released	95% Cal Maturity

Performance Data Summary

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Parameter	Unit	PALS Position	Target	Benchmark: Ashok Leyland 4220 BSVI 9S RAR 6.17	Base Vehicle: Signa 4221.T BSVI_HM RAR 5.85	Test Vehicle: Signa 4221.T 5L RDE BSVI_HM RAR 5.85	RDE Status (%) w.r.t Target	RDE Status (%) w.r.t Base	Remarks
Acceleration									
0 - 60 kmph	sec	C	≤ 79.38	79.38	79.5	**72.40**	8.79%	8.93%	
0 - 1000 m	sec	C	≤ 88.29	88.29	87.44	**84.00**	4.86%	3.93%	
Driveability									
1st Gear(4-6)	sec		Not Applicable	0.99	1.10	**1.10**	Target Not available	Target Not available	
2nd Gear(8-12)	sec			1.39	1.38	**1.49**			
3rd Gear(12-20)	sec			2.72	2.67	**2.74**			
4th Gear(19-31)	sec			4.43	4.40	**4.31**			
5th Gear(25-42)	sec			7.39	6.73	**6.48**			
6th Gear(36-50)	sec			8.94	8.31	**8.81**			
7th Gear(36-50)	sec			16.08	12.69	**12.46**			
8th Gear(36-50)	sec			37.05	31.57	**30.05**			
Rate of acceleration									
1st Gear(4-6)	kmph/sec		Not Applicable	2.03	1.82	**1.82**	Target Not available	Target Not available	
2nd Gear(8-12)	kmph/sec			2.16	2.17	**2.01**			
3rd Gear(12-20)	kmph/sec			2.21	2.25	**2.19**			
4th Gear(19-31)	kmph/sec			1.81	1.82	**1.86**			
5th Gear(25-42)	kmph/sec			1.35	1.49	**1.54**			
6th Gear(36-50)	kmph/sec			0.90	0.96	**0.91**			
7th Gear(36-50)	kmph/sec			0.62	0.79	**0.80**			
8th Gear(36-50)	kmph/sec			0.40	0.48	**0.50**			
Bath-Tub Driveability									
3rd Gear									
5-10 kmph	Sec	C	≤ 3.34	3.34	--	**3.34**	0.00%	--	
10-15 kmph	Sec	C	≤ 2.75	2.75	2.59	**1.98**	28.15%	23.71%	
4th Gear									
10-15 kmph	Sec	C	≤ 3.68	3.68	3.27	**3.13**	14.99%	4.33%	
15-20 kmph	Sec	C	≤ 2.90	2.90	2.68	**2.51**	13.56%	6.47%	
5th Gear									
10-15 kmph	Sec	C	≤ 4.79	4.79	5.54	**5.75**	-20.01%	-3.76%	
15-20 kmph	Sec	C	≤ 4.01	4.01	3.80	**3.36**	16.17%	11.54%	
20-25 kmph			≤ 3.88	3.88	4.18	**3.22**	17.01%	22.97%	
25-30 kmph	Sec	C	≤ 4.41	4.41	5.03	**4.06**	8.01%	19.35%	
6th Gear									
20-30 kmph	Sec	C	≤ 11.71	11.71	10.17	**8.75**	25.31%	14.00%	
30-40 kmph	Sec	C	≤ 11.96	11.96	12.02	**10.29**	13.94%	14.36%	
7th Gear									
20-25 kmph	Sec	C	≤ 9.33	9.33	10.61	**10.36**	-11.02%	2.37%	
25-30 kmph	Sec	C	≤ 8.34	8.34	6.50	**6.59**	21.00%	-1.36%	
30-35 kmph	Sec	C	≤ 7.97	7.97	6.78	**5.53**	30.57%	18.39%	
35-40 kmph	Sec	C	≤ 7.74	7.74	6.74	**6.24**	19.44%	7.49%	
40-45 kmph	Sec	C	≤ 7.86	7.86	7.87	**6.27**	20.19%	20.29%	
45-50 kmph	Sec	C	≤ 8.89	8.89	9.03	**7.46**	16.09%	17.39%	
8th Gear									
30-40 kmph	Sec	C	≤ 24.46	24.46	25.38	**22.64**	7.43%	10.79%	
40-50 kmph	Sec	C	≤ 25.44	25.44	21.09	**17.12**	32.70%	18.82%	
50-60 kmph	Sec	C	≤ 26.08	26.08	24.13	**20.21**	22.53%	16.27%	
60-70 kmph	Sec	C	≤ 32.36	32.36	34.12	**25.74**	20.45%	24.56%	
Gradability and Restartability									
Sustain Gradability	%	C	≥ 27.78	27.78	29.82	29.82	7.34%	0.00%	
Restart Gradability	%	C	≥ 24.26	24.26	25.72	25.72	6.02%	0.00%	

Signa 4221.T Cx BSVI RDE Performance. PALS position achieved “AL” against “AL”

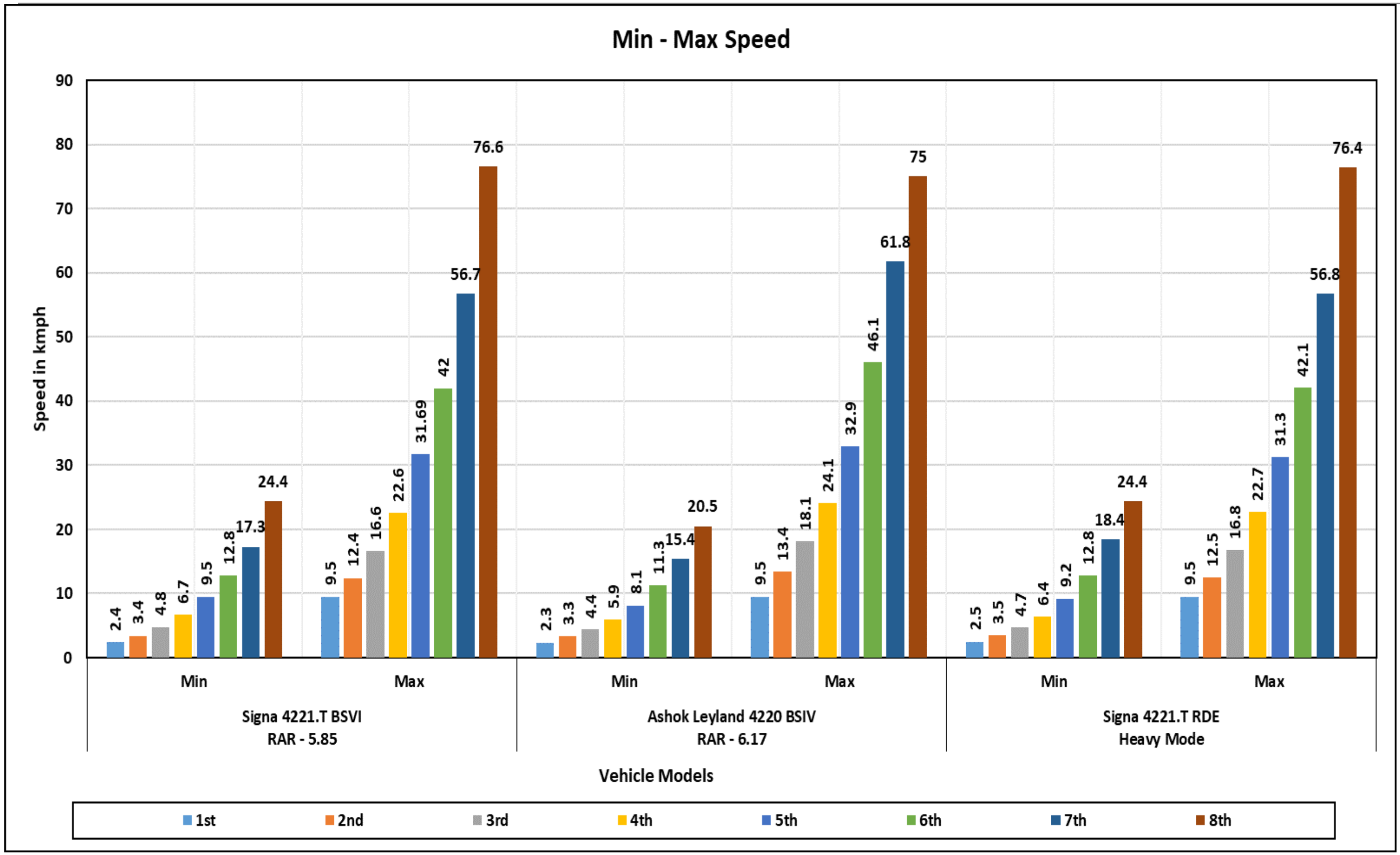
Acceleration

- ❑ Acceleration of Signa 4221.T Cx BSVI RDE was found to be **better as compared to target** in 0-60 kmph speed and 0-1000 meters band

Bath tub Drivability

- ❑ Drivability of Signa 4221.T Cx BSVI RDE was found to be **better as compared** to target in all the speed bands except 5th & 7th gear 10-15kmph & 20-25kmph speed band

Minimum & Maximum Gear Wise speed Comparison



Gear	Signa 4221.T BSVI RAR - 5.85		Ashok Leyland 4220 BSIV RAR - 6.17		Signa 4221.T RDE Heavy Mode	
	Min	Max	Min	Max	Min	Max
1st	2.4	9.5	2.3	9.5	2.5	9.5
2nd	3.4	12.4	3.3	13.4	3.5	12.5
3rd	4.8	16.6	4.4	18.1	4.7	16.8
4th	6.7	22.6	5.9	24.1	6.4	22.7
5th	9.5	31.69	8.1	32.9	9.2	31.3
6th	12.8	42	11.3	46.1	12.8	42.1
7th	17.3	56.7	15.4	61.8	18.4	56.8
8th	24.4	76.6	20.5	75	24.4	76.4

Wide open throttle Performance

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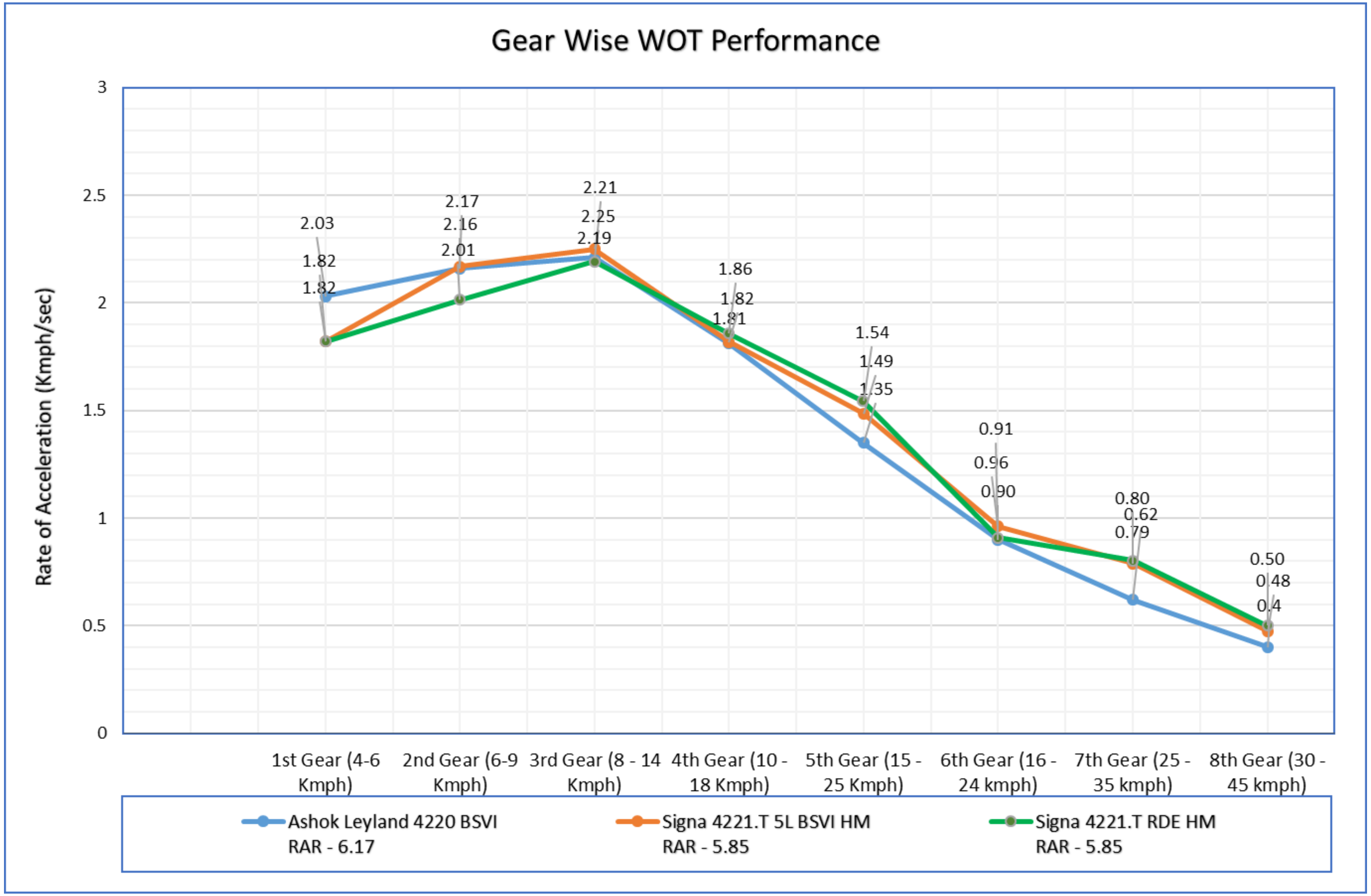
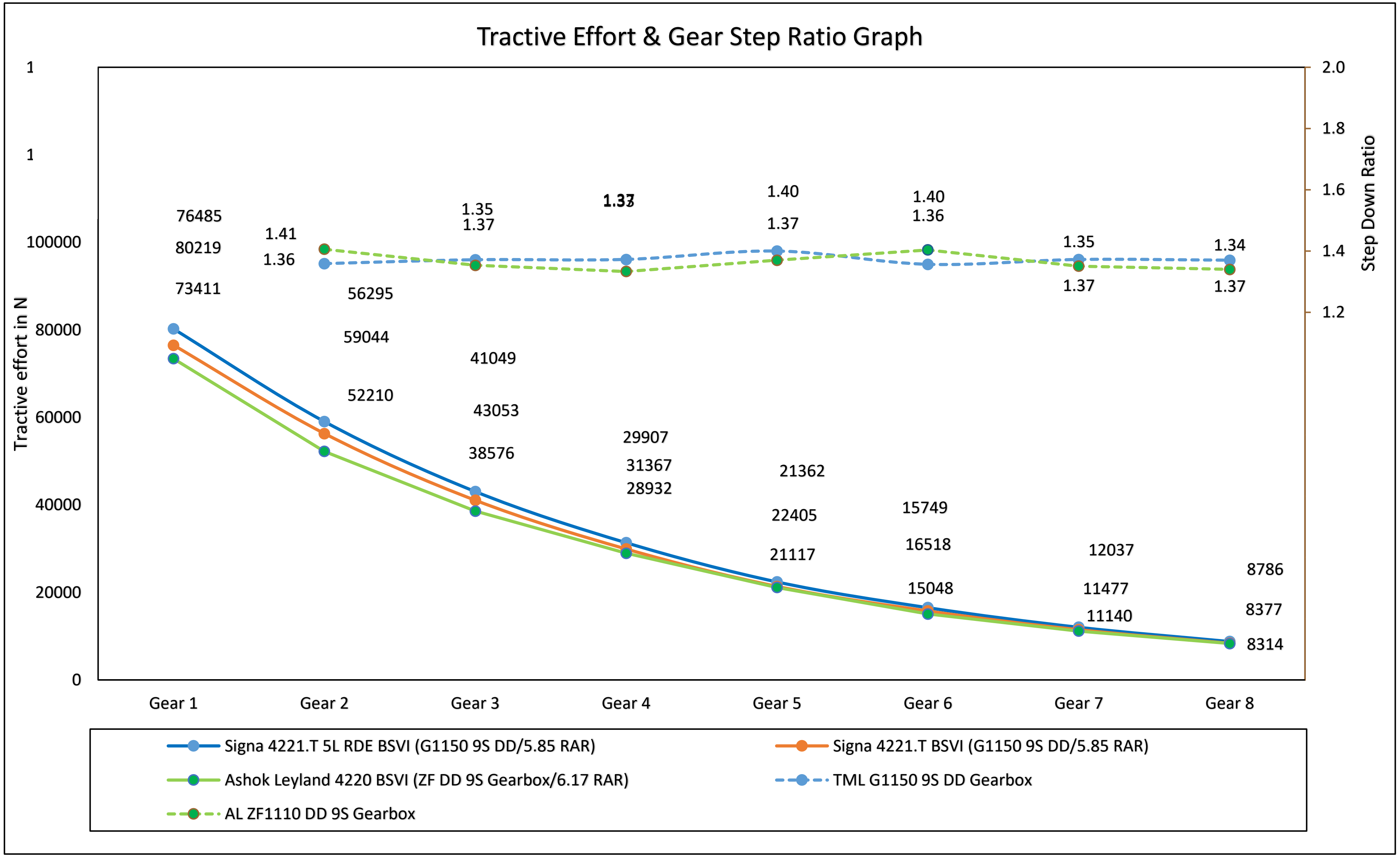


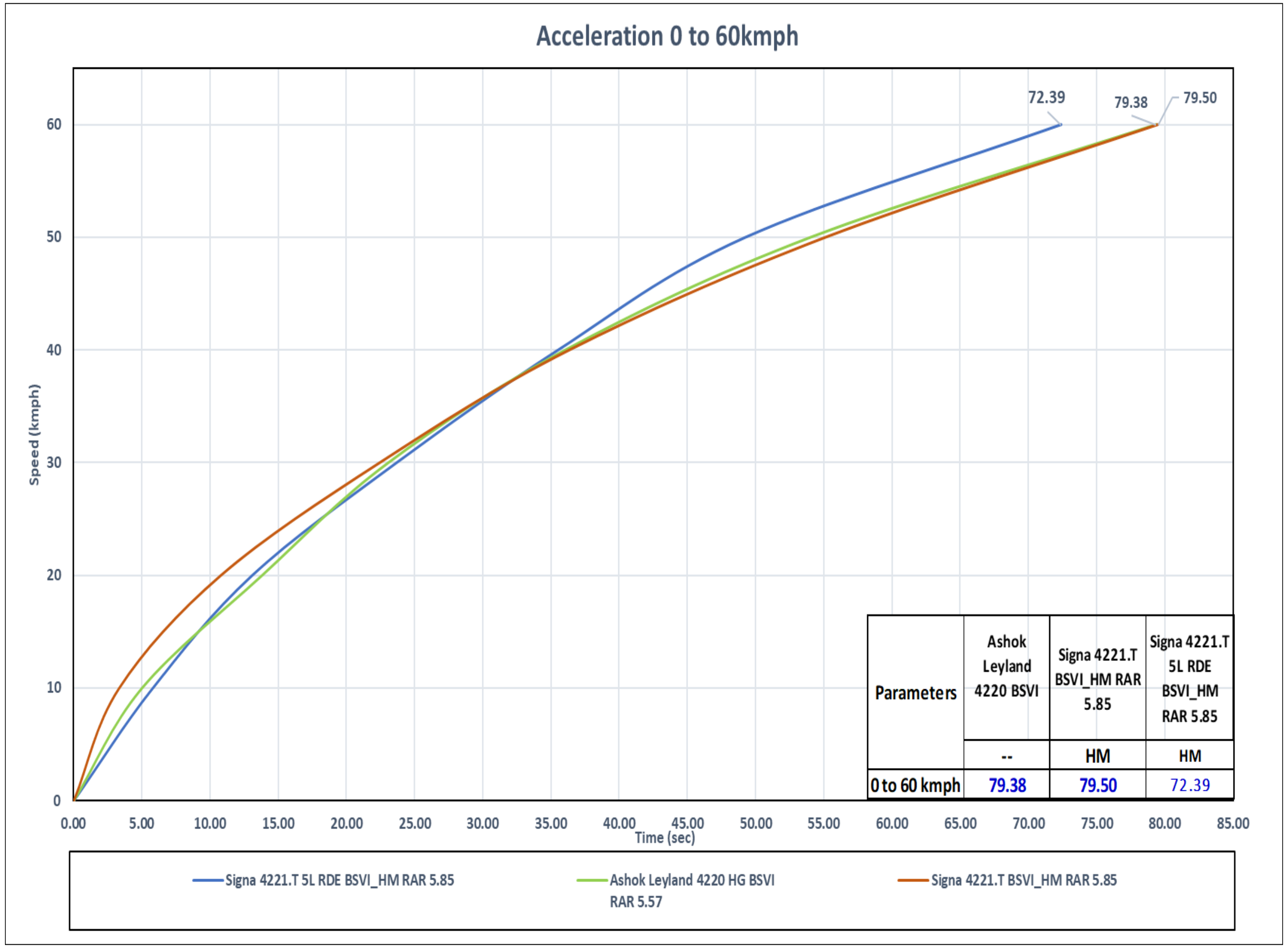
Gear	Ashok Leyland 4220 BSVI RAR - 6.17	Signa 4221.T 5L BSVI HM RAR - 5.85	Signa 4221.T RDE HM RAR - 5.85
1st Gear (4-6 Kmph)	2.03	1.82	1.82
2nd Gear (6-9 Kmph)	2.16	2.17	2.01
3rd Gear (8 - 14 Kmph)	2.21	2.25	2.19
4th Gear (10 - 18 Kmph)	1.81	1.82	1.86
5th Gear (15 - 25 Kmph)	1.35	1.49	1.54
6th Gear (16 - 24 kmph)	0.90	0.96	0.91
7th Gear (25 - 35 kmph)	0.62	0.79	0.80
8th Gear (30 - 45 kmph)	0.4	0.48	0.50

Tractive effort

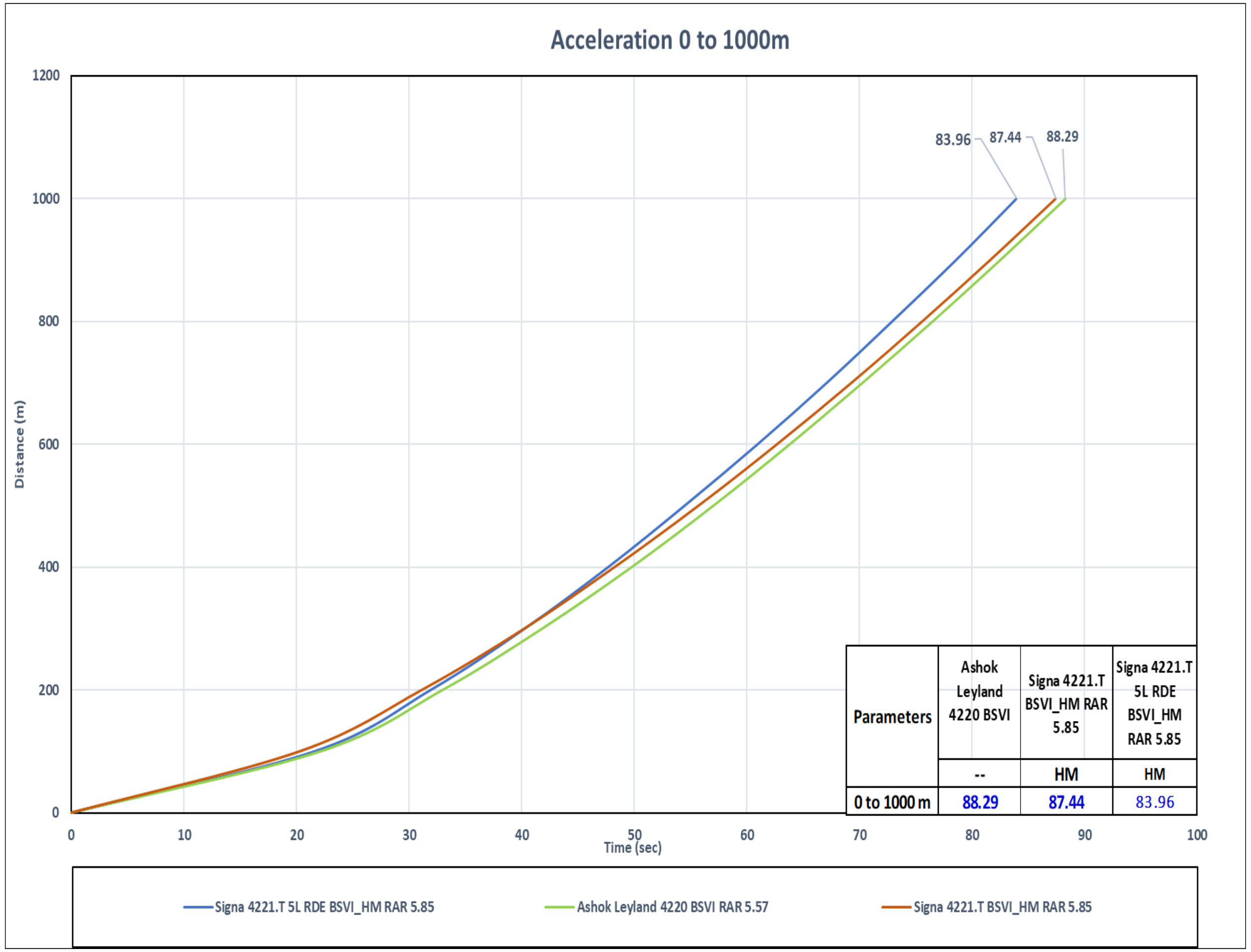


❖ Tractive effort of Signa 4221.T Cx RDE vehicle **is better as compared** with AL 4220.T BSVI & Signa 4221.T BSVI in all gear.

Acceleration Performance – Through Gear



Parameters	Ashok Leyland 4220 BSVI	Signa 4221.T BSVI_HM RAR 5.85	Signa 4221.T 5L RDE BSVI_HM RAR 5.85
	--	HM	HM
0 to 60 kmph	79.38	79.50	72.39



Parameters	Ashok Leyland 4220 BSVI	Signa 4221.T BSVI_HM RAR 5.85	Signa 4221.T 5L RDE BSVI_HM RAR 5.85
	--	HM	HM
0 to 1000 m	88.29	87.44	83.96

❖ Acceleration (0-60kmph) & (0-1000M) of Signa 4221.T Cx RDE vehicle **is better as compared** with AL 4220 & Signa 4221.T BSVI

WOT Drivability @ 1st & 2nd Gear

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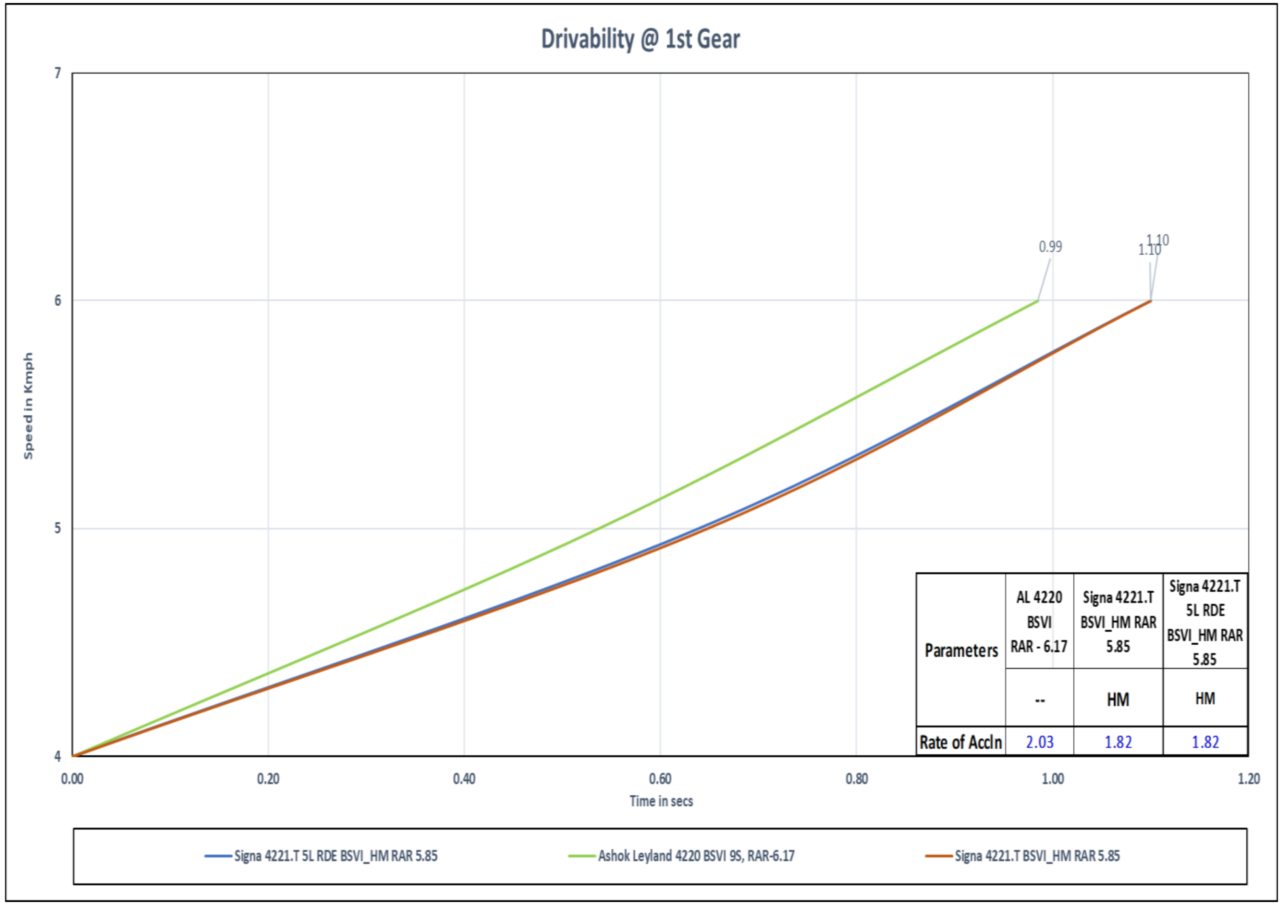
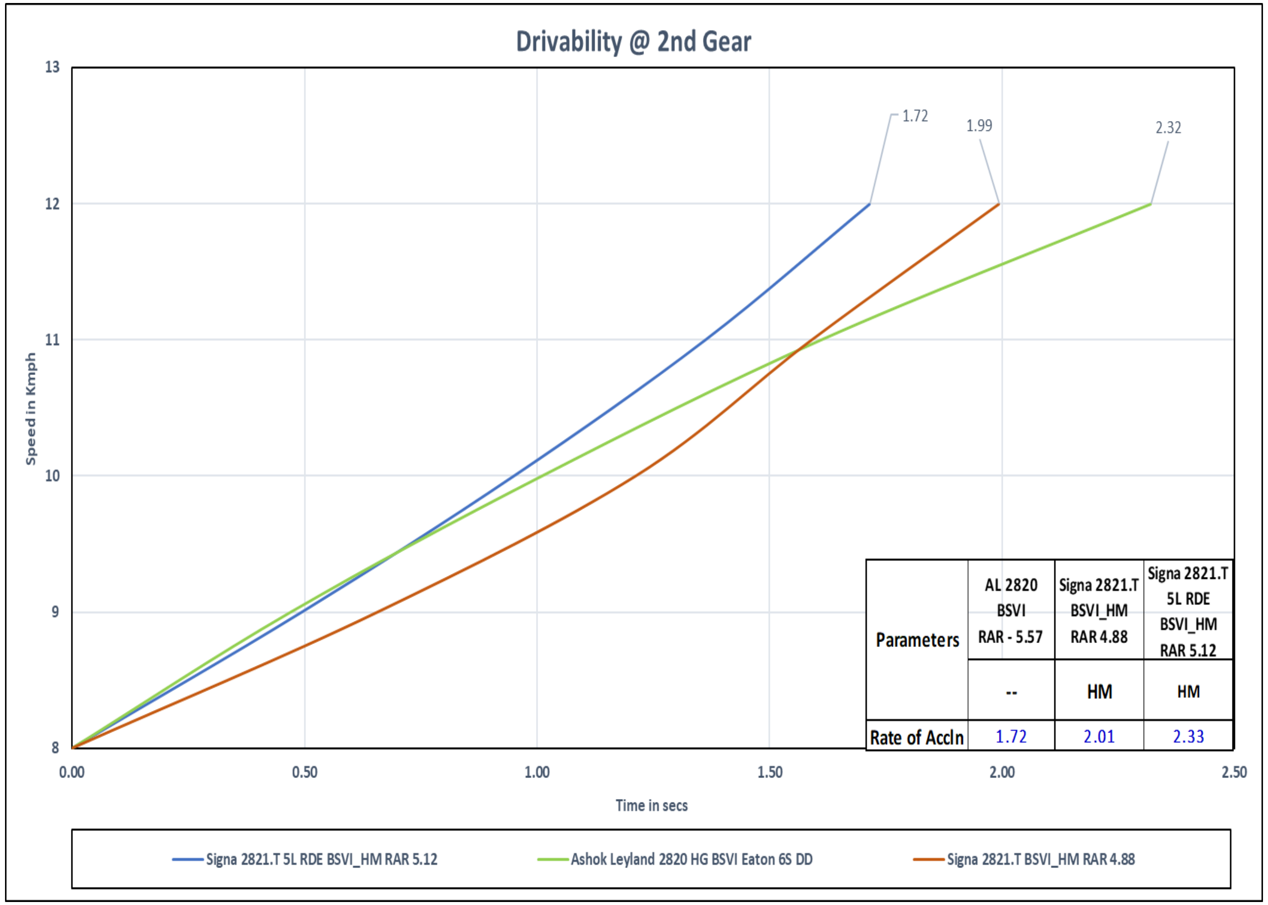


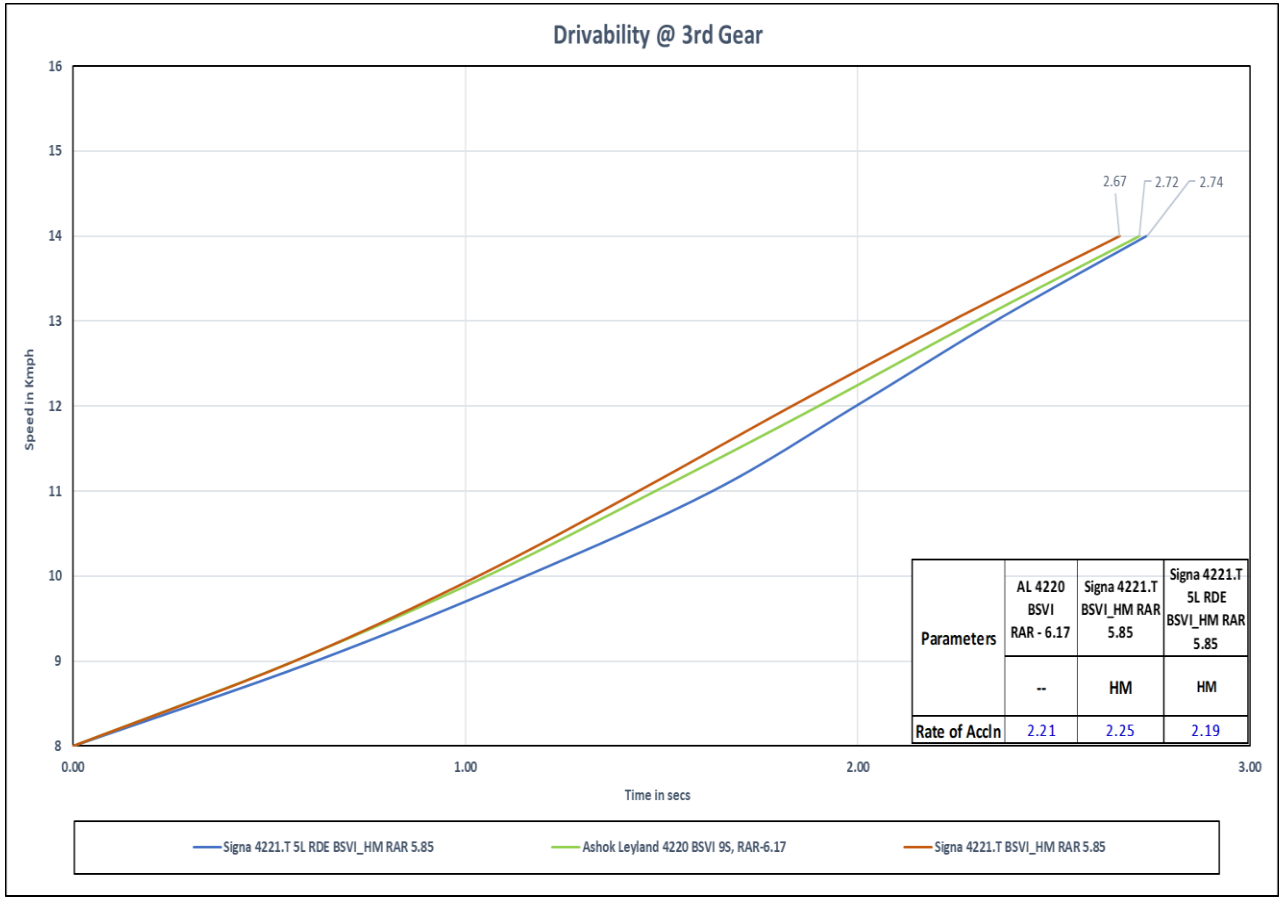
Parameters	AL 4220 BSVI RAR - 6.17	Signa 4221.T BSVI_HM RAR 5.85	Signa 4221.T 5L RDE BSVI_HM RAR 5.85
	--	HM	HM
Rate of Accln	2.03	1.82	1.82



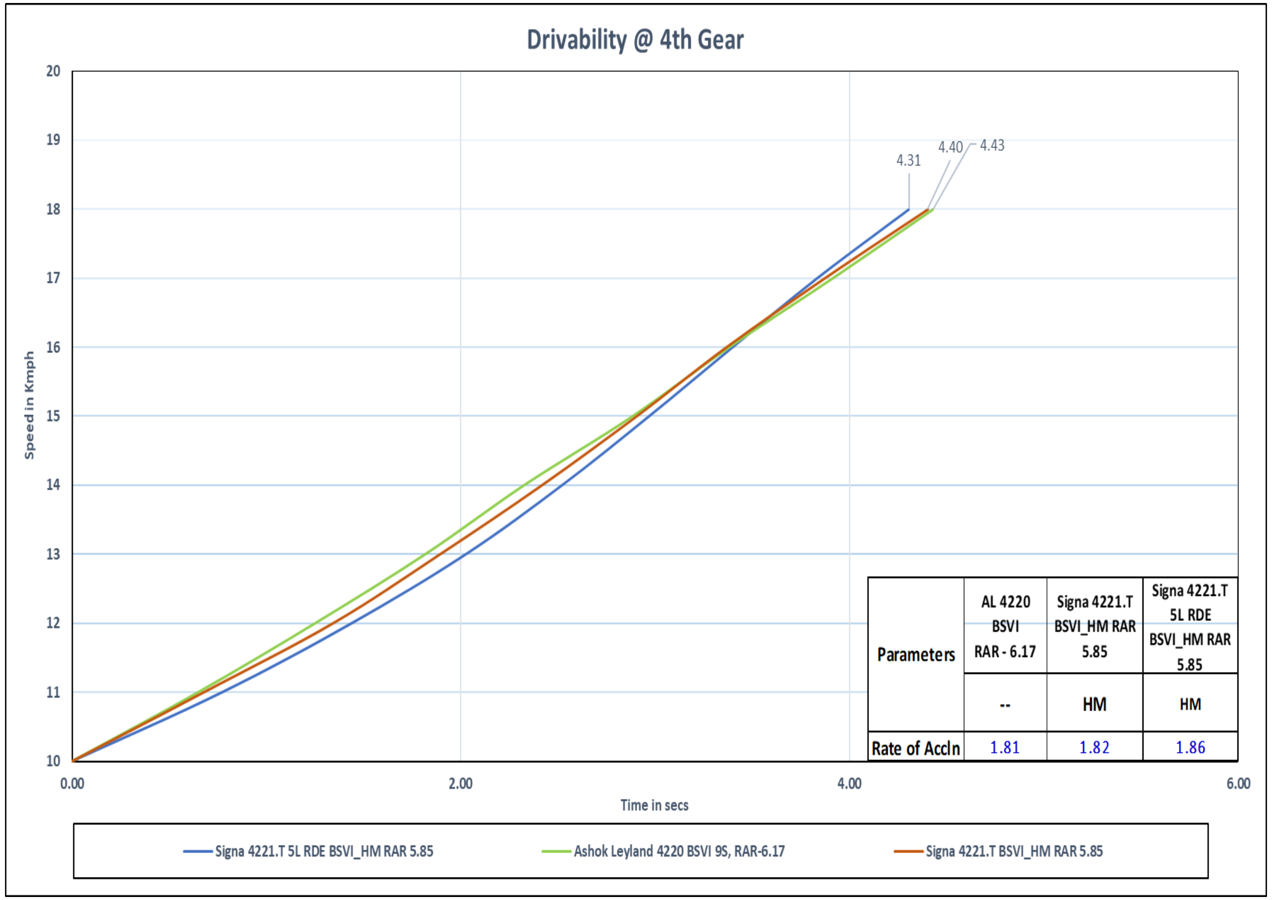
Parameters	AL 2820 BSVI RAR - 5.57	Signa 2821.T BSVI_HM RAR 4.88	Signa 2821.T 5L RDE BSVI_HM RAR 5.12
	--	HM	HM
Rate of Accln	1.72	2.01	2.33

- In 1st gear, WOT performance of Signa 4221.T Cx RDE vehicle **is Inferior** but in 2nd gear it's **better** as compared with AL 4220 & Signa 4221.T BSVI Veh.

WOT Drivability @ 3^{rd} & 4^{th} Gear



Parameters	AL 4220 BSVI RAR - 6.17	Signa 4221.T BSVI_HM RAR 5.85	Signa 4221.T 5L RDE BSVI_HM RAR 5.85
	--	HM	HM
Rate of Accln	2.21	2.25	2.19



Parameters	AL 4220 BSVI RAR - 6.17	Signa 4221.T BSVI_HM RAR 5.85	Signa 4221.T 5L RDE BSVI_HM RAR 5.85
	--	HM	HM
Rate of Accln	1.81	1.82	1.86

- In 3^{rd} gear, WOT acceleration performance of Signa 4221.T Cx RDE vehicle **is inferior** as compared with AL 4220 & Signa 4221.T BSVI
- In 4^{th} gear, WOT acceleration performance of Signa 4221.T Cx RDE vehicle **is marginally better** as compared with AL 4220.T BSVI & Signa 4221.T BSVI

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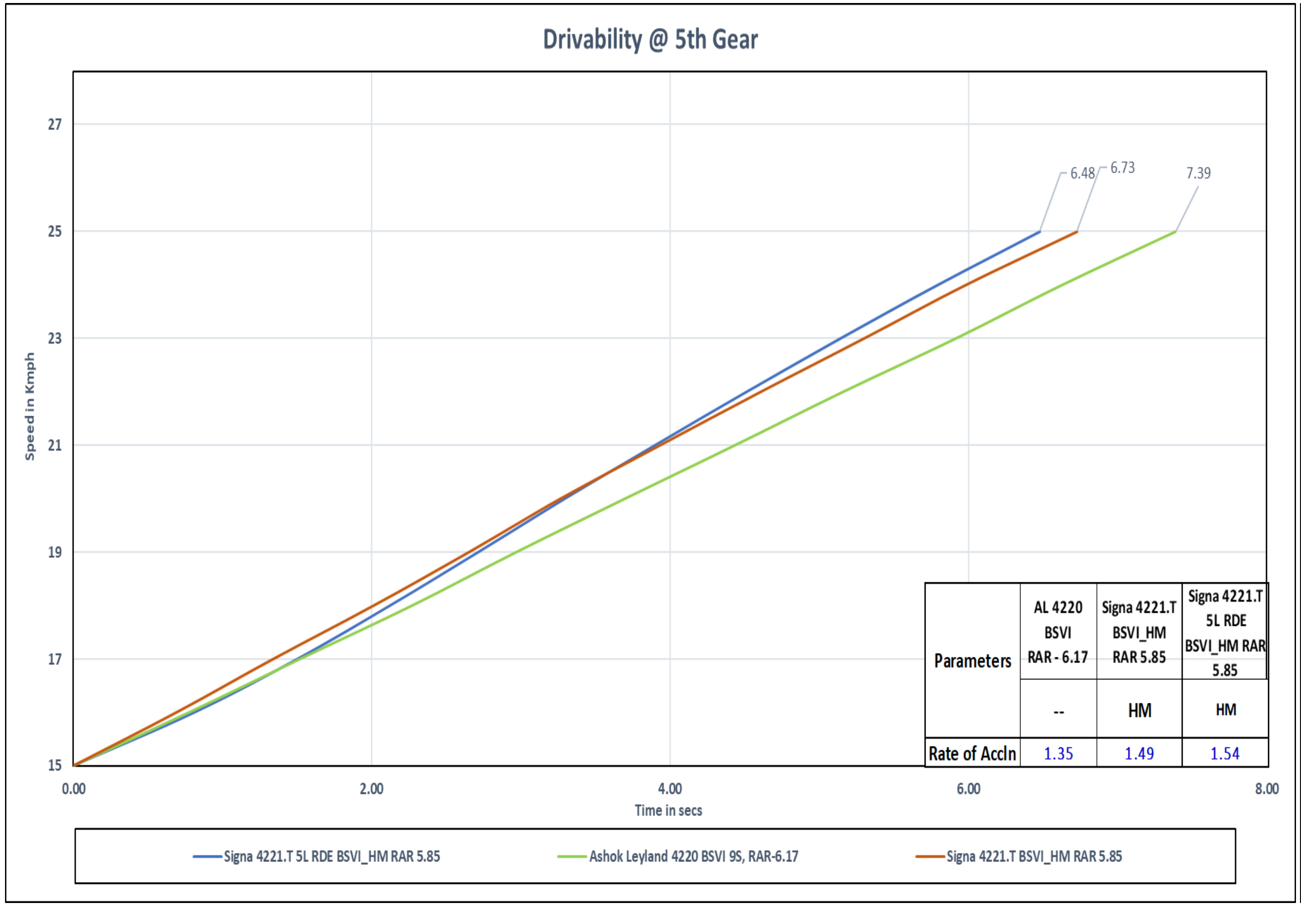
WOT Drivability @ 5th & 6th Gear



Parameters	AL 4220 BSVI RAR - 6.17	Signa 4221.T BSVI_HM RAR 5.85	Signa 4221.T 5L RDE BSVI_HM RAR 5.85
	--	HM	HM
Rate of Accln	1.35	1.49	1.54

Parameters	AL 4220 BSVI RAR - 6.17	Signa 4221.T BSVI_HM RAR 5.85	Signa 4221.T 5L RDE BSVI_HM RAR 5.85
	--	HM	HM
Rate of Accln	0.90	0.96	0.91

- ❖ In 5th gear, WOT acceleration performance of Signa 4221.T Cx RDE vehicle **is better as compared** with AL 4220 & Base vehicle
- ❖ In 6th gear, WOT acceleration performance of Signa 4221.T Cx RDE vehicle **is inferior as compared** with Base vehicle.

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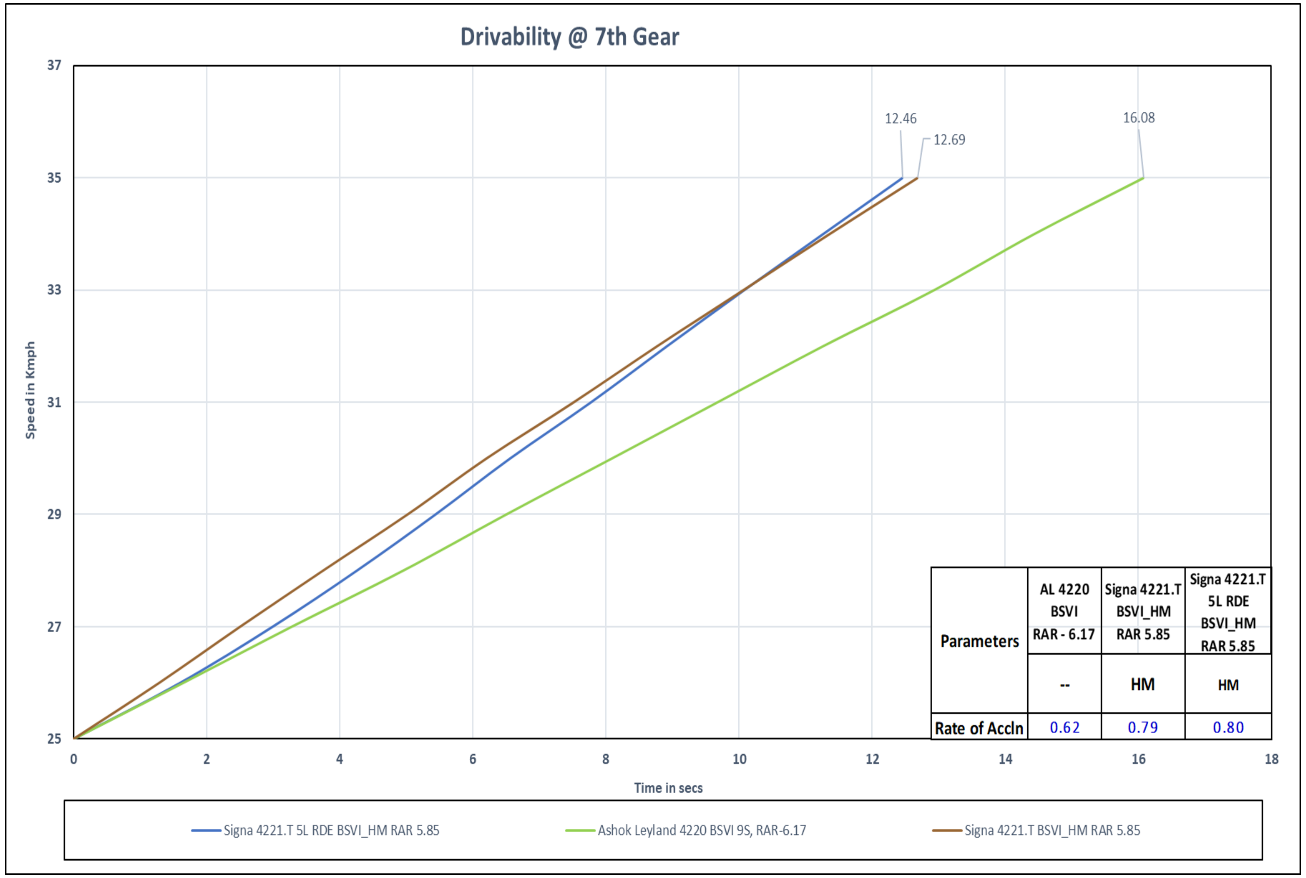
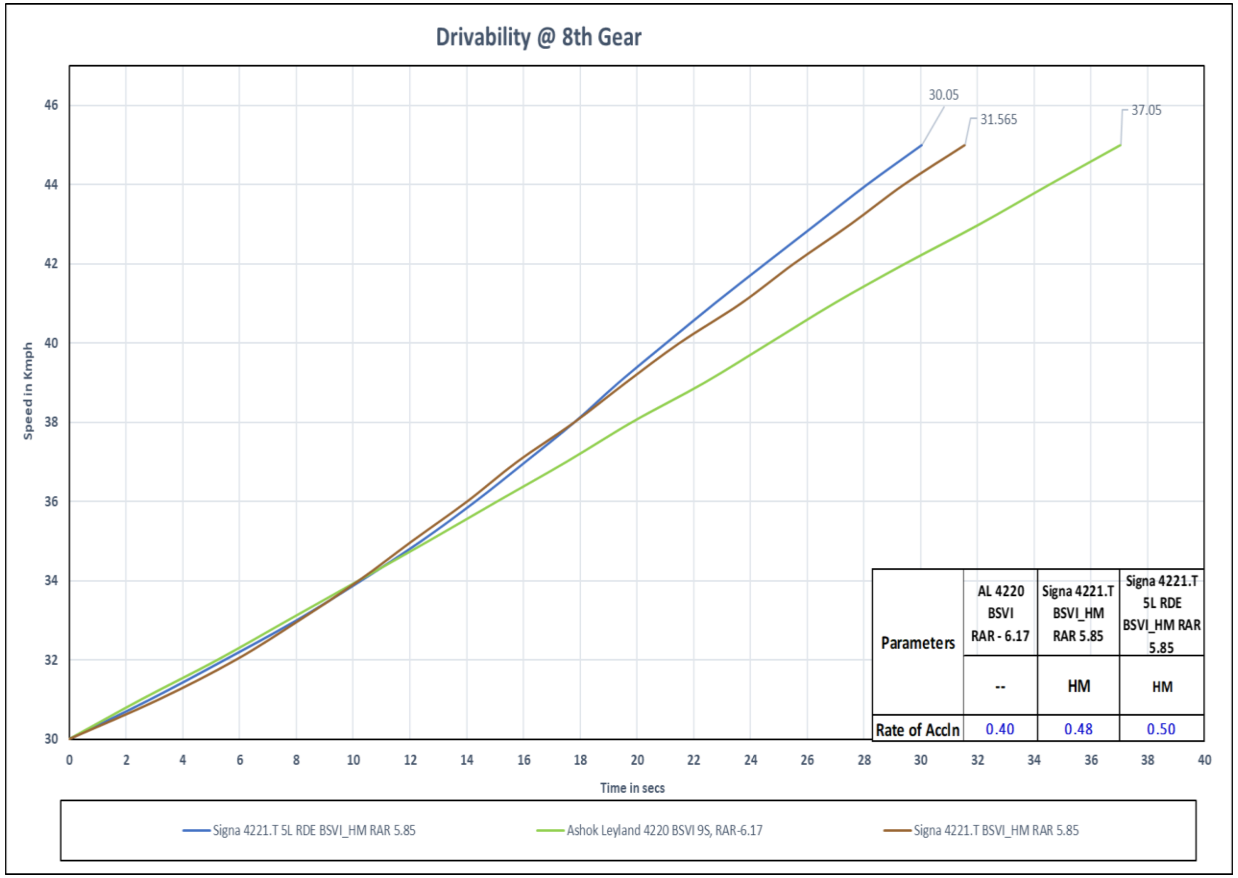
WOT Drivability @ 7th & 8th Gear





- In 7th gear, WOT acceleration performance of Signa 4221.T Cx RDE vehicle **is better** as compared with AL 4220 BSVI vehicle.
- In 8th gear, WOT acceleration performance of Signa 4221.T Cx RDE vehicle **is better as compared** with AL 4220 & Base vehicle.

Bathtub Drivability Report

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Gear	Speed band	Ashok Leyland 4220 BSVI RAR 6.17	Signa 4221.T BSVI_HM RAR 5.85	Signa 4221.T 5L RDE BSVI_HM RAR 5.85
		Time in Sec	Time in Sec	Time in Sec
3rd Gear	5-10 kmph	3.34		3.34
	10-15 kmph	2.75	2.59	1.98
4th Gear	10-15 kmph	3.68	3.27	3.13
	15-20 kmph	2.90	2.68	2.51
5th Gear	10-15 kmph	4.79	5.54	5.75
	15-20 kmph	4.01	3.80	3.36
	20-25 kmph	3.88	4.18	3.22
	25-30 kmph	4.41	5.03	4.06
6th Gear	20-30 kmph	11.71	10.17	8.75
	30-40 kmph	11.96	12.02	10.29
7th gear	20-25 kmph	9.33	10.61	10.36
	25-30 kmph	8.34	6.50	6.59
	30-35 kmph	7.97	6.78	5.53
	35-40 kmph	7.74	6.74	6.24
	40-45 kmph	7.86	7.87	6.27
	45-50 kmph	8.89	9.03	7.46
8th gear	30-40 kmph	24.46	25.38	22.64
	40-50 kmph	25.44	21.09	17.12
	50-60 kmph	26.08	24.13	20.21
	60-70 kmph	32.36	34.12	25.74

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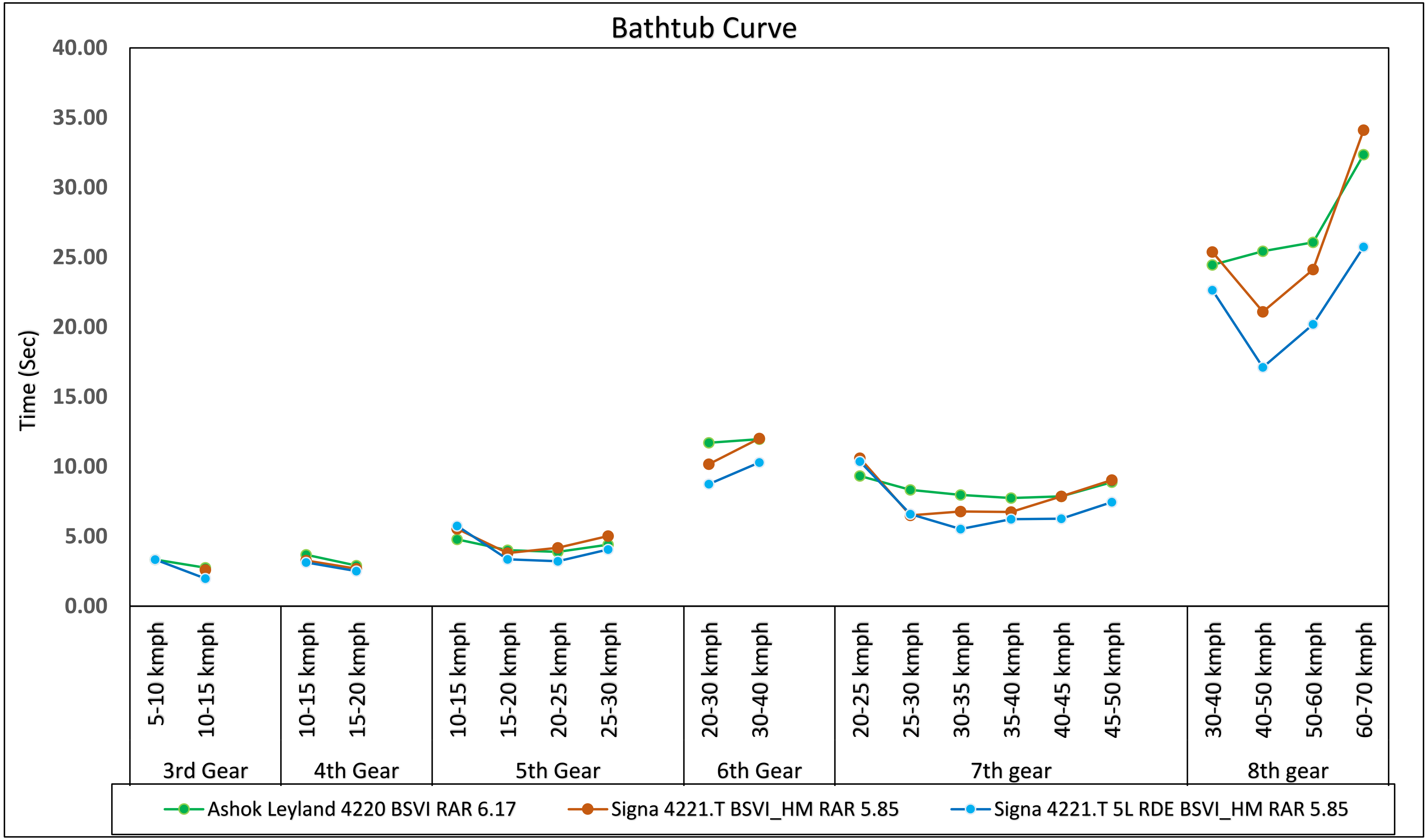
Bathtub Drivability Comparison with Benchmark & Base Vehicles



❖ Bathtub Drivability of Signa 4221.T Cx RDE vehicle **is better** as compared with AL 4220 BSVI & Base vehicle in all Gears.

Coast Down Comparison with Benchmark & Base Vehicles

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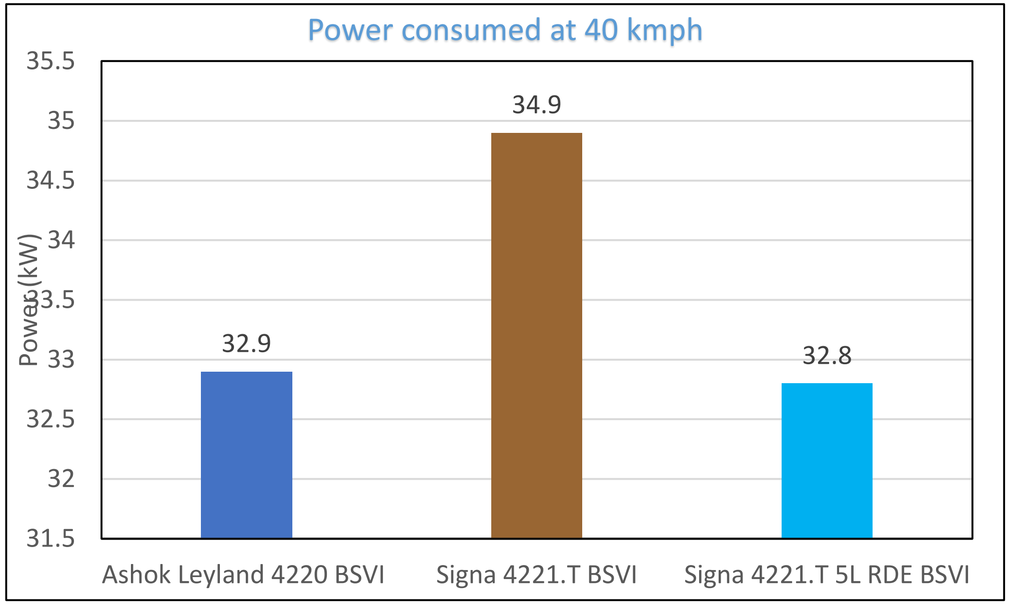
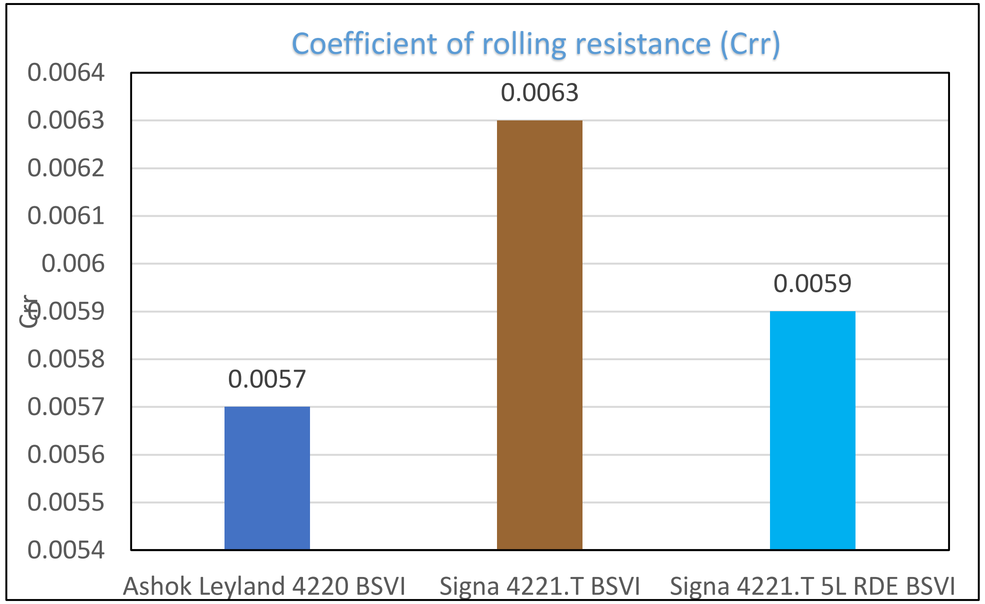
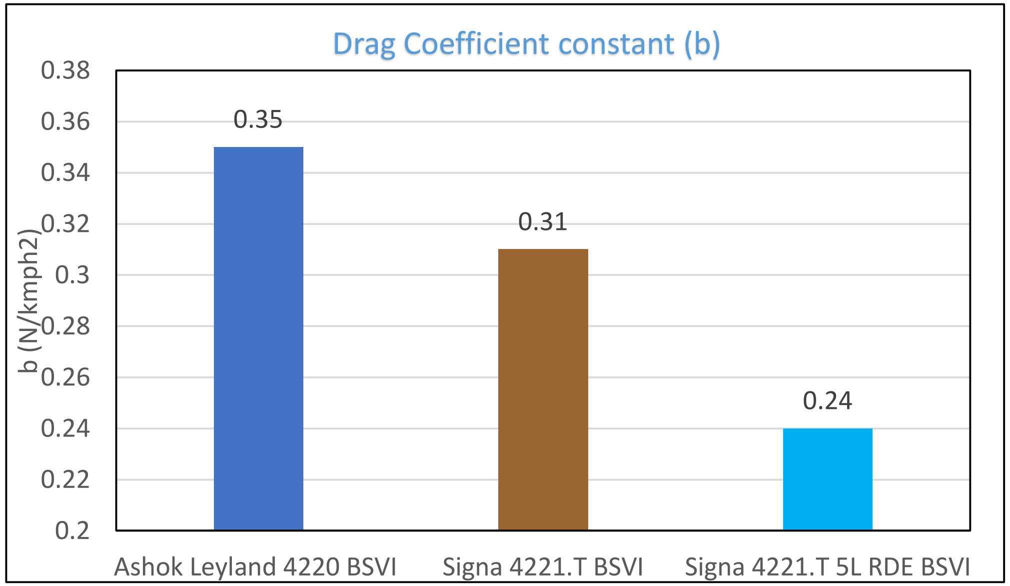
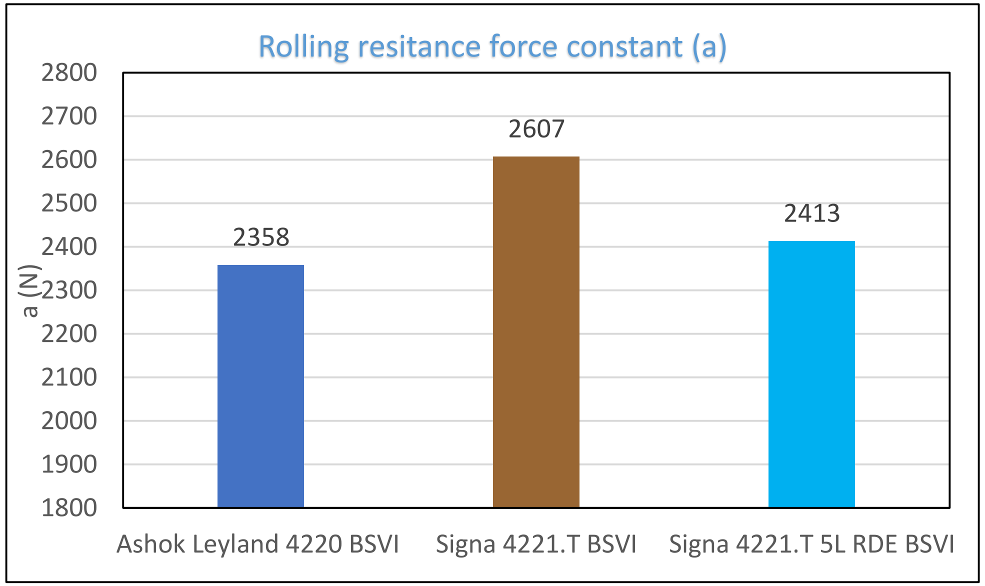
Vehicle	Power consumed at 40 kmph	a	b	Coeff of rolling resistance
	kW	N	N/kmph^2	--
Ashok Leyland 4220 BSVI	32.9	2358	0.35	0.0057
Signa 4221.T BSVI	34.9	2607	0.31	0.0063
Signa 4221.T 5L RDE BSVI	32.8	2413	0.24	0.0059



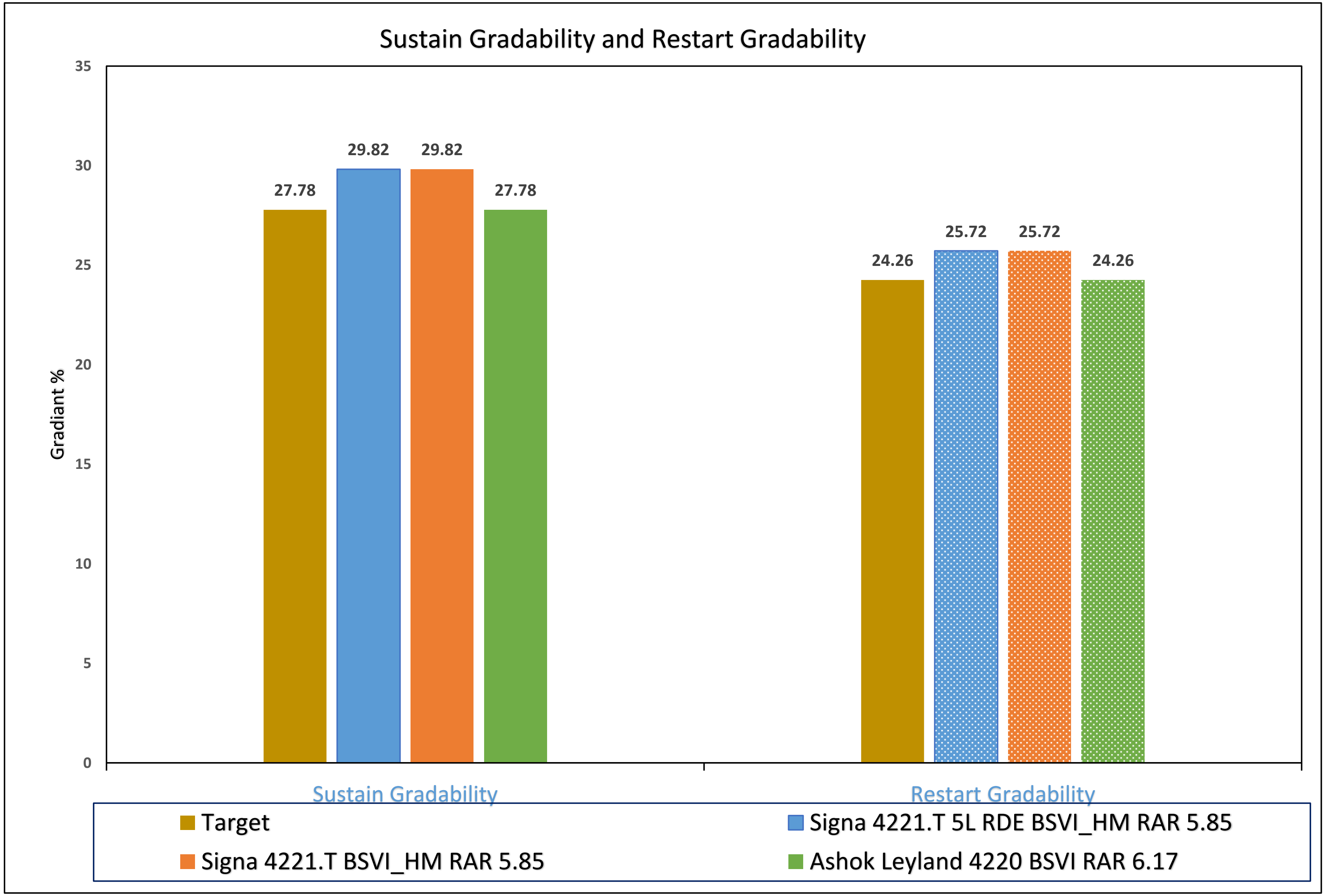






Sustained and Restart Gradeability

TATA MOTORS
Connecting Aspirations



Test	Sustain Gradability and Restart Gradability			
	Target	Ashok Leyland 4220 BSVI RAR 6.17	Signa 4221.T BSVI_HM RAR 5.85	**Signa 4221.T 5L RDE BSVI_HM RAR 5.85**
Sustain Gradability	27.78	27.78	29.82	29.82*
Restart Gradability	24.26	24.26	25.72	25.72*

Thank You!

TEST REQUISITION

Ref.	22404168R/PD/RDE/Signa 4221.T	Date:	23/06/2023

To,
ERC Testing
Kindly arrange to take up for testing of the following component.

1	Part Description	TATA SIGNA 4221.T 5L BSVI				
2	Part Number	22404168R,				
3	Drawing number	224000000018,	Mod status	NR,1,	Mod Date	17/07/2023
4	Vehicle Model / Project	LPT 3721 5L BS4 SIGNA WITH AC RA109RR 10R20				
5	WBS Number	P.21.EA.458 -TRUCK 4425 / 4930 RDE				
6	Supplier	NA				
7	Number of samples	--				
8	Inspection report	Enclosed	☐ Yes	☑ No	If No commitment date:	
9	Material report	Enclosed	☐ Yes	☑ No	If No commitment date:	
10	Suppliers test report	Enclosed	☑ Yes	☐ No	If No commitment date:	
11	Reason for testing	First Source Development	☐	Weight Reduction	☐	
		Alternate source development	☐	Cost Reduction	☐	
		Data generation Test	☑	Regulatory test	☐	
		Any other (pl. specify)	☐	New Development	☐	
12	What testing required?	DYNAMIC				
13	Remarks	PD-Performance and Driveability : 22404168R/PD/RDE/Signa 4221.T				
14	Peripheral Parts Availability	NO				

Authorized Signature:	
Name:	
Dept:	

Contact Person:	
Phone Number	

For Testing Dept Use only	Test Item Number and Date(stamp)	Test Engineer

TATA MOTORS LIMITED ERC PUNE	TEST PLAN VEHICLE TESTING DEPARTMENT	**DATE:** 19/07/2023			
Description: Performance evaluation of Signa 4221.T Cx RDE in comparison with Ashok Leyland 4220.T BSVI and Signa		**Test Item No:** 42546			
		New	✔	**Moderate**	x
		Major	x	**Minor**	x

Part No: 22404168R, **Drawing No:** 224000000018,	**Modification No:** NR,1,	**Enclosures:** PD Trials of Signa 4221.T 5L RDE BSVI Cx Vehicle
Make: NA	**WBS Number:** P.21.EA.458 -TRUCK 4425 / 4930 RDE	**Copies to** :
Vehicle Type: LPT 3721 5L BS4 SIGNA WITH AC RA109RR 10R20	**No. of Samples:** 1	
Test Requested by:	VEHICLE-I	**Equipments:**
Test Requisition Reference:	Testing-Outdoor/15688	
Reason for Test/s:	Data Generation	
Regulatory Requirement:	a) Component:	
	b) Vehicle:	

Sr. No.	Test description	Test specification	Acceptance criteria	No. of Samples	Test Responsibility	Proto Phase (Mule/ Alpha/ Beta/PP)	Remarks
1	Performance & Driveability tests	As per PED standards	As per PALS	1	Palash Das	Beta	NA

REMARKS: Test plan for test execution	**INTERNAL REF:**	
TEST TO FAILURE: No		
MEOST: No		
RELIABILITY TESTING: No		
BENCHMARKING TESTING: No		
DFMEA/FMEA DONE: Yes		
PREPARED BY	**CHECKED BY**	**APPROVED BY**